

Development of the Korean Benthic Macroinvertebrate Sediment Index (KBSI) for assessing sediment-associated ecological condition in freshwater ecosystems

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Abstract

Freshwater sediments accumulate organic matter, nutrients, metals, and fine particles and can influence benthic macroinvertebrate assemblages. However, biological tools for evaluating sediment-associated ecological condition remain limited in Korean freshwater systems. This study developed the Korean Benthic Macroinvertebrate Sediment Index (KBSI) as a field-based biological indicator using data from 80 sites and 320 site-survey sample units. Sediment variables, including porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), the Metals Tier II Index, and mud fraction, were summarized using principal component analysis. PC1 represented an integrated sediment-condition gradient associated with organic enrichment, reducing conditions, relative sediment-bound metal contamination, and fine-particle accumulation. PC1 scores from 309 sample units containing selected indicator taxa were divided into nine class intervals using a Weibull cumulative distribution function and Sturges' rule. Taxon-specific tolerance values and indicator weights were derived from occurrence distributions along this gradient and used to calculate KBSI. KBSI showed significant negative relationships with TAN, AVS, TOC, the Metals Tier II Index, mud fraction, and PC1, indicating that lower index values were associated with more degraded sediment-associated field conditions. The relationship between observed and calculated KBSI values supported internal agreement within the development dataset but did not constitute independent validation. KBSI values were classified into five ecological condition classes from A to E for practical interpretation. KBSI is intended for biological assessment of depositional sediment habitats and should be interpreted together with sediment physicochemical and habitat information rather than as an independent chemical sediment-quality standard or direct toxicological endpoint.

Keywords: Benthic macroinvertebrates; Biological assessment; Freshwater sediment; KBSI;

40 Principal component analysis; Sediment-associated ecological condition

41

1. Introduction

Sediment in rivers and lakes is a particulate matrix of gravel, sand, silt, clay, and organic matter that stores watershed-derived pollutants over extended periods and exchanges solutes with the water column as a dynamic reactive system (Chapman, 1990; Zhou et al., 2024). Settling and resuspension, sorption and desorption, and redox-mediated transformations continually modify sediment properties (El-Sharkawy et al., 2025; Hiki et al., 2023). Where fine fractions dominate, contaminant accumulation and bioavailability increase and can restructure benthic macroinvertebrate assemblages (Relyea et al., 2012; Wharton et al., 2023). Therefore, single-time measurements of water-column concentrations do not adequately represent cumulative contamination or habitat alteration, underscoring the need for biological indicators that reflect sediment-associated ecological conditions.

The Oligochaete Index of Sediment Bioindication (IOBS) uses oligochaete assemblage composition to evaluate sediment-associated biological condition in relation to sediment impairment (AFNOR, 2002; AFNOR, 2016), and the U.S. Equilibrium Partitioning Sediment Benchmarks (ESB) quantify linkages between sediment chemistry and biological effects (U.S.EPA, 2003). The Sediment Quality Triad (SQT) integrates chemistry, toxicity testing, and community data to diagnose ecological degradation in contaminated sediments (Chapman, 1990). Stress-specific biotic indices have also been developed: the Fine Sediment Biotic Index (FSBI) encodes sensitivities to fine particles (Relyea et al., 2012), and the Proportion of Sediment-sensitive Invertebrates (PSI) uses the share of sensitive taxa to evaluate restoration outcomes (Extence et al., 2010).

In coastal and estuarine environments, the Benthic Response Index (BRI) and the AZTI Marine Biotic Index (AMBI) apply weighted-tolerance principles and have informed analogous approaches for fresh waters (Borja et al., 2000; Smith et al., 2001). More recently,

multivariate formulations—such as the Hellenic Lake Littoral Biotic Index (HeLLBI) and the Cumulative Biotic Index (CBI)—have related community structure to pollution gradients within explicit statistical frameworks, while high-throughput metabarcoding is improving congruence with morphology-based indices (Lobo et al., 2017; Mavromati et al., 2021; Sudarso et al., 2026).

In Korea, nationally standardized tools for water-column conditions—such as the Korean Benthic macroinvertebrate Index of Biotic Integrity (KB-IBI)—are established, but sediment-focused indicators remain comparatively underdeveloped (Jun et al., 2012). A domestic prototype, the Benthic macroinvertebrate Sediment Index (BSI), estimated taxon tolerance values and indicator weights from relative frequencies in freshwater sediments and computed a weighted mean using a Sediment Quality Index that integrates total organic carbon (TOC), ammonia ($\text{NH}_3\text{-N}$), and heavy metals (Kim et al., 2019). Although BSI showed statistical associations with sediment impairment gradients, its generality was constrained by limited sample size and taxonomic uncertainty within Chironomidae, indicating the need for a tool calibrated with broader spatial coverage and refined taxonomy.

To address this gap, we developed the Korean Benthic Macroinvertebrate Sediment Index (KBSI) as a field-based biological indicator of sediment-associated ecological condition in Korean freshwaters. Specifically, this study aimed to: (1) assemble a nationwide dataset of co-located benthic macroinvertebrate assemblages and sediment physicochemical conditions from major Korean river basins and representative waterbody types; (2) define an integrated sediment environmental gradient using PCA of porewater total ammonia nitrogen, acid-volatile sulfides, total organic carbon, heavy metals, and the mud fraction; (3) derive taxon-specific tolerance values and indicator weights from macroinvertebrate occurrence patterns along the PC1-based sediment gradient; and (4) formulate and examine KBSI as a community-level

index by relating index values to individual sediment variables, the PC1 sediment gradient, and observed–calculated index agreement. KBSI is therefore intended not as a stand-alone chemical or toxicological measure of sediment quality, but as a biological indicator of benthic assemblage responses to sediment-associated environmental conditions under field settings.

2. Materials and methods

2.1 Study area and sampling design

2.1.1 Site description

This study was conducted at 80 freshwater sediment monitoring sites distributed across the major river basins of Korea (Fig. 1). The sites were selected to represent broad gradients in sedimentary environments, waterbody type, and anthropogenic influence, including both lotic and lentic systems. For descriptive and management-oriented interpretation, the sites were categorized into five groups: general streams, urban streams, industrial streams, lakes, and reservoirs. The stream categories were assigned according to the dominant surrounding land-use setting and the original monitoring-site designation, rather than using uniform quantitative catchment-scale land-use thresholds. Urban streams represented lotic sites located in predominantly urbanized settings, whereas industrial streams represented sites associated with industrial areas. General streams comprised the remaining lotic sites that were not classified as urban or industrial streams.

These categories were used to describe broad differences in waterbody type, surrounding land-use setting, and sediment depositional environment, rather than to define strict reference or impairment classes. General streams were therefore not treated as reference or best-available sites and could include streams influenced by agriculture, forestry, flow

113 regulation, impoundments, or other catchment and hydromorphological pressures.

114 Field surveys were conducted repeatedly between 2022 and 2024, producing a total of
115 320 site-survey sample units. For the derivation of taxon-specific tolerance values and indicator
116 weights, 309 sample units in which selected indicator taxa occurred were used. Sample units
117 without selected indicator taxa were excluded from the tolerance-value calculation because
118 they did not provide the taxon-occurrence information required to estimate biological responses
119 along the sediment environmental gradient.

120 The sampling design was developed to support a field-based biological assessment of
121 sediment-associated ecological condition. Accordingly, biological and sediment samples were
122 collected from comparable depositional habitats within each site, where fine sediments and
123 associated organic or reduced materials could accumulate. The sampling framework was
124 therefore intended for depositional sediment habitats, not for coarse riffle or boulder-dominated
125 substrates. This design allowed the KBSI to characterize benthic macroinvertebrate assemblage
126 responses to sediment-associated environmental conditions under field settings.

128 *2.1.2. Collection and Identification of Benthic Macroinvertebrates*

129 Benthic macroinvertebrates were collected from depositional sediment habitats using
130 a cylindrical core sampler with an internal diameter of 7.5 cm. At each site, five replicate cores
131 were taken to a depth of 20 cm where coring was feasible (Kim et al., 2018; Song et al., 2025).
132 The 20 cm sampling depth was adopted to capture not only organisms concentrated in the upper
133 sediment layer, but also burrowing taxa, particularly oligochaetes, that may occur in deeper
134 sediment layers.

135 In the field, sediment core contents were washed through a 0.20-mm sieve to minimize

the loss of small-bodied sediment-dwelling organisms, particularly oligochaetes and chironomid larvae. Retained materials were preserved in ethanol and transported to the laboratory. The use of a 0.20-mm sieve was selected because small-bodied taxa in fine sediments can be underestimated when larger mesh sizes are used (Song et al., 2025).

In the laboratory, residual sediment was removed and organisms were sorted under a stereomicroscope. Specimens were identified to the lowest practicable taxonomic level, generally to species where diagnostic characters were available, using standard taxonomic references (Cummins and Merritt, 1996; Kawai and Tanida, 2005; Peckarsky et al., 1990; Yoon, 1988). Chironomid larvae were identified based on diagnostic structures such as the head capsule, mentum, mandible, and antennae (Kwak, 2015). Oligochaetes were identified using external and reproductive morphological characters, including chaetal and reproductive structures (Park et al., 2016). All identified specimens were preserved in 80% ethanol to minimize morphological deformation.

Taxa that were sufficiently frequent and taxonomically reliable were retained as candidate indicator taxa for KBSI development. This taxonomic screening was necessary because the derivation of tolerance values and indicator weights requires reliable occurrence information along the sediment environmental gradient.

2.1.3. Sediment Sampling and Physicochemical Analysis

Surface sediment samples for physicochemical analysis were collected concurrently with the biological samples from the corresponding depositional habitats within each site using a cylindrical core sampler with an internal diameter of 7.5 cm. For sediment chemistry and particle-size analysis, the upper 0–7 cm sediment layer was used because this layer represents

the biologically active surface sediment where organic matter accumulation, redox gradients, porewater chemistry, and contaminant binding are most pronounced. This analytical depth differed from the macroinvertebrate sampling depth of 20 cm, which was used to include deeper-burrowing taxa.

Each sediment sample was sealed in a 300-mL high-density polyethylene bottle, stored at ≤ 4 °C, and transported to the laboratory. The target sediment variables were porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), heavy metals, and particle-size composition. TAN was used as an indicator of nitrogenous organic enrichment in sediment porewater, AVS represented reduced sediment conditions associated with sulfide formation, and TOC represented organic matter accumulation.

TAN was quantified colorimetrically at 630 nm. TOC was measured using a CHNS analyzer after removal of inorganic carbon. AVS was determined as Ag₂S precipitated under a nitrogen stream, followed by filtration and gravimetric drying. For heavy metals, sediment samples were digested with mixed acids and analyzed using ICP–AES or ICP–MS; Hg was quantified using TDA–AAS. Relative sediment-bound metal contamination was summarized using the Metals Tier II Index based on the Tier II sediment guideline values for Cu, Pb, Ni, As, Hg, Zn, Cd, and Cr (NIER, 2022).

$$\text{Metals Tier II Index} = \frac{\sum_{i=1}^8 \left(\frac{EC_i}{PELK_i} \right)}{8} \quad (1)$$

Where EC_i is the measured concentration of metal i and $PELK_i$ is the corresponding Tier II guideline value. Higher values indicate greater relative sediment-bound metal contamination compared with the Tier II sediment guideline benchmarks.

Particle-size fractions were separated using 2-mm and 75- μ m sieves into gravel (>2 mm), sand (0.075–2 mm), and silt–clay (<0.075 mm) fractions. In this study, the fine-particle

variable used for KBSI development refers to the silt–clay fraction, hereafter termed the mud fraction, and does not include sand. The mud fraction was treated as a physical depositional variable rather than a chemical contaminant, because fine particles can co-vary with organic matter accumulation, reduced conditions, and contaminant retention in depositional habitats. Routine quality assurance and quality control procedures followed the national sediment monitoring guidelines (NIER, 2022).

2.2. Development and calculation of the Korean Benthic Macroinvertebrate Sediment Index (KBSI)

2.2.1. PCA-derived sediment environmental gradient

To derive an integrated sediment environmental gradient for KBSI development, principal component analysis (PCA) was performed using the sediment physicochemical variables measured at each sample unit. PCA was used to reduce redundancy among correlated environmental variables and to summarize the dominant multivariate gradient in sediment condition (Abdi and Williams, 2010; Jolliffe and Cadima, 2016). The variables included porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), the Metals Tier II Index, and the mud fraction. These variables were selected to represent major sediment-associated conditions, including nitrogenous organic enrichment, reduced sediment conditions, organic matter accumulation, relative sediment-bound metal contamination, and fine-particle deposition.

Prior to PCA, variables with skewed distributions were transformed to reduce the influence of extreme values. TAN, AVS, TOC, and the Metals Tier II Index were log-transformed, whereas the mud fraction, expressed as proportional data, was arcsine square-root

transformed. All transformed variables were standardized to a mean of zero and a standard deviation of one before analysis, and PCA was conducted on the standardized environmental matrix (Legendre and Legendre, 2012; Quinn and Keough, 2002).

The first principal component score was used as the integrated sediment environmental gradient for deriving taxon-specific tolerance values and indicator weights. PC1 was selected because it represented the dominant axis of variation in sediment conditions and provided a continuous gradient along which macroinvertebrate occurrence patterns could be evaluated. In this study, PC1 was not interpreted as a stand-alone chemical pollution index or an independent measure of sediment toxicity. Instead, it was used as a multivariate sediment-condition gradient reflecting field-based co-variation among sediment physicochemical variables.

The PC1 scores of sample units containing selected indicator taxa were used in the subsequent derivation of biological indicator values. Among the 320 site-survey sample units, 309 sample units with selected indicator taxa were retained for this step, whereas sample units without selected indicator taxa were excluded because they did not provide occurrence information required for estimating taxon responses along the sediment environmental gradient.

2.2.2. Classification of the PC1 gradient and occurrence distribution

To classify the PC1-based sediment environmental gradient for deriving biological indicator values, the distribution of PC1 scores was analyzed using sample units in which selected indicator taxa occurred. Among the 320 site-survey sample units, 309 sample units were retained for this step after excluding 11 sample units in which no selected indicator taxa were recorded.

The distribution of PC1 scores for the 309 sample units was fitted using a Weibull

cumulative distribution function, which was selected because of its flexibility in describing skewed environmental-gradient distributions and deriving probability-based class boundaries (Weibull, 1951). The Weibull cumulative distribution function was expressed as:

$$F(x) = 1 - \exp\left[-\left(\frac{x-\theta}{\lambda}\right)^k\right] \quad (2)$$

where $F(x)$ is the cumulative probability at PC1 score x , θ is the location parameter, λ is the scale parameter, and k is the shape parameter.

The number of PC1 class intervals was determined using Sturges' rule, resulting in nine class intervals (Sturges, 1926):

$$K = 1 + \log_2(n) \quad (3)$$

where K is the number of class intervals and n is the number of sample units. For the 309 sample units used in this step, Sturges' rule produced approximately nine intervals.

Class boundaries were obtained from the inverse Weibull cumulative distribution function so that each PC1 interval represented an equal probability range:

$$x_p = \theta + \lambda[-\ln(1-p)]^{1/k} \quad (4)$$

where x_p is the PC1 score corresponding to cumulative probability p . Each interval was assigned a representative class value, which was subsequently used to summarize the occurrence distribution of each indicator taxon along the PC1 gradient. This procedure provided a standardized basis for estimating taxon-specific tolerance values and indicator weights from field occurrence data.

In this framework, the PC1 classes were not treated as regulatory sediment-quality standards. Rather, they were used as empirical intervals of the multivariate sediment-condition gradient to quantify biological occurrence patterns. This interpretation is important because

PC1 includes both chemical variables and a physical depositional variable, the mud fraction, which may reflect natural hydromorphological processes as well as anthropogenic sediment accumulation.

2.2.3. Derivation of taxon-specific tolerance values and indicator weights

Taxon-specific tolerance values and indicator weights were derived from the occurrence distribution of selected indicator taxa along the PC1-based sediment environmental gradient. Only taxa with reliable taxonomic resolution and sufficient occurrence frequency were retained as indicator taxa, because rare or taxonomically uncertain taxa may produce unstable tolerance estimates. This screening was intended to ensure that KBSI was based on taxa with interpretable field occurrence patterns along the sediment-condition gradient. This occurrence-based approach follows the general logic of deriving biological indicator values from taxon distributions along environmental gradients (Kim et al., 2019; Kong and Kim, 2016). The final indicator set therefore represents a screened subset of taxa suitable for KBSI calibration rather than a complete inventory of all taxa collected during the surveys.

For each indicator taxon, its occurrence frequency was calculated within each of the nine PC1 class intervals. The occurrence frequency of taxon i in class k was expressed as:

$$RF_{ik} = \frac{n_{ik}}{N_k} \times 100 \quad (5)$$

where RF_{ik} is the relative occurrence frequency of taxon i in PC1 class k , n_{ik} is the number of sample units in class k where taxon i occurred, and N_k is the total number of sample units in class k .

The occurrence frequencies of each taxon across the nine PC1 classes were then

normalized to obtain its occurrence valency:

$$V_{ik} = \frac{RF_{ik}}{\sum_{k=1}^9 RF_{ik}} \quad (6)$$

where V_{ik} represents the proportional occurrence valency of taxon i in class k . Taxon-specific tolerance values were calculated as the weighted average of the PC1 class values using occurrence valencies as weights:

$$t_i = \sum_{k=1}^9 C_k V_{ik} \quad (7)$$

where t_i is the tolerance value of taxon i , and C_k is the representative class value assigned to PC1 class k . Higher tolerance values indicate that a taxon occurred more frequently under sediment conditions associated with higher PC1-derived stress levels, whereas lower values indicate occurrence concentrated under less degraded sediment conditions.

Indicator weights were assigned according to the concentration of each taxon's normalized occurrence valencies across the nine PC1 classes. Taxa whose occurrence valencies were concentrated within a limited portion of the PC1 gradient were assigned higher indicator weights, whereas taxa with occurrence valencies distributed broadly across multiple classes were assigned lower weights. Thus, the indicator weight represents the specificity of a taxon's occurrence distribution along the sediment-condition gradient, rather than its tolerance position or physiological sensitivity.

This distinction between tolerance value and indicator weight prevents taxa with similar tolerance positions but different occurrence breadths from being treated as equivalent. In the final indicator table, taxa with strongly concentrated occurrence distributions were assigned higher indicator weights, whereas taxa occurring across broader portions of the PC1 gradient were assigned lower weights.

2.2.4. Calculation and classification of the Korean Benthic Macroinvertebrate Sediment Index (KBSI)

The Korean Benthic Macroinvertebrate Sediment Index (KBSI) was calculated as a weighted community-level index using the taxon-specific tolerance value, frequency value, and indicator weight of each indicator taxon. For each sample unit, taxa included in the final indicator list were matched with their corresponding tolerance values and indicator weights, and their frequency values were used to represent their relative contribution to the assemblage. This weighted-index structure follows the general approach of combining taxon tolerance, occurrence, and indicator strength in benthic macroinvertebrate bioassessment (Kim et al., 2019).

The KBSI was calculated as:

$$KBSI = (10 - \frac{\sum_{i=1}^n t_i h_i g_i}{\sum_{i=1}^n h_i g_i}) \times 10 \quad (8)$$

where n is the number of indicator taxa occurring in a sample unit, t_i is the tolerance value of taxon i , h_i is the frequency value of taxon i , and g_i is the indicator weight of taxon i .

The weighted mean tolerance value was subtracted from 10 so that higher KBSI values indicate better sediment-associated ecological condition, whereas lower values indicate assemblages dominated by taxa tolerant of degraded sediment conditions.

KBSI values were scaled from 0 to 100 and classified into five ecological condition classes. For practical interpretation, KBSI values were classified into five assessment classes following the KBSI assessment framework developed in this study: A, high, 70–100; B, good,

60–<70; C, moderate, 50–<60; D, poor, 40–<50; and E, bad, 0–<40.

These classes provide interpretive categories for biological assessment of sediment-associated ecological condition. They should not be interpreted as independent chemical sediment-quality standards, regulatory criteria, or toxicological thresholds. Rather, they summarize KBSI values within the assessment framework developed for this study.

3. Results and discussion

3.1. PCA-derived sediment environmental gradient

Principal component analysis summarized the covariance structure of the five sediment physicochemical variables used to characterize sediment-associated field conditions: porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), Metals Tier II Index, and mud fraction. The first three PCA axes explained 86.0% of the total variance, with PC1 accounting for 58.0%, PC2 for 17.2%, and PC3 for 10.8% (Table 1; Fig. 2). PC1 was the dominant axis because all five sediment variables showed clear associations with this axis. The loadings on PC1 were 0.66 for ln(TAN), 0.69 for ln(AVS), 0.86 for ln(TOC), 0.82 for ln(Metals Tier II Index), and 0.76 for ln(mud fraction). Thus, PC1 represented a composite sediment environmental gradient along which nitrogenous enrichment, reducing sediment conditions, organic matter accumulation, relative sediment-bound metal contamination, and fine-particle deposition co-varied.

Organic matter accumulation is ecologically relevant because it can increase oxygen demand in surface sediments and promote conditions unfavorable to oxygen-sensitive benthic fauna (Rosenberg et al., 2001). AVS reflects reduced sediment conditions, and sulfide-rich or strongly reduced sediments can modify metal binding, porewater chemistry, and exposure

conditions for sediment-dwelling organisms (Simpson et al., 2013). TAN was interpreted as a porewater indicator of nitrogenous organic enrichment rather than as a stand-alone toxicant gradient, because ammonia effects in sediments depend on pH, redox condition, and porewater–overlying water exchange. The Metals Tier II Index summarized the relative degree of sediment-bound metal contamination, but its ecological interpretation should consider bioavailability and site-specific geochemical conditions rather than total concentration alone (Simpson et al., 2013). Therefore, the strong contribution of TOC, AVS, TAN, and the Metals Tier II Index indicates that PC1 captured a biogeochemical gradient associated with organic enrichment, reduced sediment conditions, and relative sediment-bound metal contamination.

Mud fraction also contributed strongly to PC1, but its interpretation differs from that of chemical contaminants. Fine particles can reduce interstitial flow, alter oxygen exchange, and modify benthic habitat structure in lotic environments (Wood and Armitage, 1997). Fine-grained sediment can also affect macroinvertebrate assemblages by filling substrate interstices and reducing the availability of coarse-substrate microhabitats used by rheophilic or sediment-sensitive taxa (Bylak and Kukuła, 2022). In addition, fine sediments can enhance the retention of organic matter and particle-reactive contaminants because of their large surface area and depositional stability (Wharton et al., 2023). However, mud accumulation may arise from natural hydromorphological and depositional processes as well as from anthropogenic changes in flow regulation, land use, and sediment supply (Wharton et al., 2023; Wood and Armitage, 1997). Accordingly, mud fraction was treated as a physical depositional variable rather than as a chemical contaminant.

For these reasons, PC1 was interpreted as a field-based multivariate sediment-condition gradient, not as a stand-alone sediment toxicity index or a regulatory sediment-quality standard. Sediment chemistry and benthic community composition can provide

complementary lines of evidence for sediment-associated ecological assessment, but neither line of evidence alone fully resolves causal mechanisms under field conditions (Long and Chapman, 1985). The sediment quality triad framework similarly emphasizes that sediment chemistry, toxicity, and benthic community responses should be interpreted together rather than treated as interchangeable measures (Chapman, 1990). In the present study, KBSI was derived from macroinvertebrate occurrence patterns along the PC1 gradient; therefore, it should be interpreted as a biological indicator of sediment-associated ecological condition under field settings, not as an independent chemical or toxicological assessment of sediment quality.

3.2. Classification of PC1 scores and derivation of class values

Among the 320 site-survey sample units, 309 sample units contained selected indicator taxa and were therefore used to classify the PC1 sediment-condition gradient. The remaining 11 sample units were excluded from this step because they did not contain selected indicator taxa and therefore provided no taxon-occurrence information for deriving tolerance values or indicator weights. The PC1 scores of the 309 retained sample units were used to describe the empirical distribution of sediment-associated field conditions represented by the indicator-taxon dataset.

The distribution of PC1 scores was fitted using a Weibull cumulative distribution function (Fig. 3). The Weibull model was used to describe the probability distribution of PC1 scores and to derive class boundaries along the continuous sediment-condition gradient. Because the objective was to construct empirical intervals for biological indicator derivation rather than to define regulatory sediment-quality thresholds, the distribution was divided into equal cumulative-probability intervals. Sturges' rule produced nine class intervals for the 309 sample units, and the inverse Weibull cumulative distribution function was used to estimate the

PC1 boundary and representative class value for each interval (Table 2).

The resulting PC1 classes covered the observed sediment-condition gradient from low PC1 scores to high PC1 scores. Under the sign convention used in this study, higher PC1 scores represented stronger sediment-associated environmental stress, including greater organic enrichment, reducing conditions, relative sediment-bound metal contamination, and mud accumulation. The representative class values therefore provided a standardized numerical basis for estimating taxon-specific occurrence positions along the sediment-condition gradient. Taxa occurring mainly in lower PC1 classes were assigned lower tolerance values, whereas taxa occurring mainly in higher PC1 classes were assigned higher tolerance values.

Importantly, these PC1 classes should not be interpreted as sediment-quality standards or toxicity thresholds. They are empirical intervals of a field-based multivariate sediment-condition gradient derived from the distribution of PC1 scores in sample units containing indicator taxa. This distinction is necessary because PC1 integrates chemical variables with mud fraction, a physical depositional variable that can reflect both natural hydromorphological processes and anthropogenic sediment accumulation. The class values were therefore used only as an internal scaling framework for deriving KBSI tolerance values and indicator weights, not as independent criteria for classifying sediment contamination or toxicity.

3.3. Taxon-specific tolerance values and indicator weights

Taxon-specific tolerance values and indicator weights were derived from the occurrence distributions of selected indicator taxa across the nine PC1 class intervals (Table 3). In total, 137 indicator taxa were retained in the final indicator table. This approach follows the logic of field-based macroinvertebrate bioassessment, in which taxon scores are inferred from

observed distributions along environmental gradients rather than from a single chemical variable or laboratory endpoint (Hering et al., 2004; Rosenberg and Resh, 1993). It also builds on the saprobic valency concept and its later adaptations, where taxa are characterized by both their position along an environmental gradient and their strength of indication within that gradient (Sládeček, 1967; Yoon et al., 1992; Zelinka and Marvan, 1961).

Although KBSI shares a weighted-indicator structure with the Korean BMI (Kong et al., 2018) and conventional saprobic indices, the indices differ in their target environmental gradients and application domains. BMI is used primarily to assess the general ecological condition of Korean streams, whereas KBSI was derived from depositional sediment habitats in both lotic and lentic systems. Similarly, conventional saprobic values primarily represent taxon distributions along gradients of organic pollution and oxygen conditions, whereas KBSI tolerance values were independently derived along a multivariate sediment-condition gradient integrating TAN, AVS, TOC, relative metal contamination, and mud fraction. Correspondence among taxon classifications may therefore reflect shared responses to organic enrichment and oxygen depletion, but KBSI tolerance values should not be interpreted as conventional saprobic values.

Tolerance values represent the weighted average position of each taxon along the PC1 sediment-condition gradient. Lower values indicate taxa whose occurrences were concentrated in lower PC1 classes, whereas higher values indicate taxa whose occurrences were concentrated in higher PC1 classes. Under the sign convention used in this study, higher PC1 scores represented stronger sediment-associated environmental stress, including greater organic enrichment, reducing conditions, metal contamination, and mud accumulation. Accordingly, tolerance values should be interpreted as field-derived occurrence positions along the measured PC1 gradient, not as fixed physiological toxicity thresholds.

The resulting values showed broad taxon-level variation. Taxa with low tolerance values included *Rhyacodrilus coccineus* ($t = 0.0$), *Chaetogaster limnaei* ($t = 0.0$), *Limnodrilus claparedeianus* ($t = 0.7$), and *Chaetogaster diaphanus* ($t = 1.9$), whereas taxa with higher tolerance values included *Limnodrilus udekemianus* ($t = 7.3$) and *Branchiodrilus hortensis* ($t = 7.1$). These differences are consistent with the general observation that benthic macroinvertebrate assemblages vary along gradients of organic enrichment, oxygen depletion, sediment condition, and habitat disturbance (Townsend et al., 2008; Von Bertrab et al., 2013). However, the values should not be over-interpreted as universal taxon traits because field distributions can also reflect habitat structure, flow regime, sampling depth, and regional species pools.

Indicator weights describe a different property from tolerance values. A high indicator weight means that a taxon's occurrences were concentrated within a restricted portion of the PC1 gradient, whereas a low weight indicates a broader distribution across classes. Thus, indicator weight represents occurrence specificity rather than sensitivity or tolerance itself. This distinction follows the weighted-indicator logic of saprobic and Korean benthic macroinvertebrate indices, where community-level scores combine taxon position, abundance or occurrence, and indicator strength (Zelinka & Marvan, 1961; Yoon et al., 1992; Kong & Kim, 2016; Kim et al., 2019).

This distinction also resolves the issue of taxa with tolerance values of 0.0. In the final table, *Rhyacodrilus coccineus* and *Chaetogaster limnaei* had tolerance values of 0.0 and indicator weights of 5, because their occurrences were concentrated in the lowest PC1 class. Their high indicator weights therefore reflect narrow occurrence specificity, not an automatic assignment based on sensitivity. Conversely, taxa with intermediate or high tolerance values may receive low weights if they are broadly distributed across PC1 classes. This treatment

prevents tolerance value and indicator weight from being conflated into a single sensitivity metric.

Overall, Table 3 should be interpreted as a field-derived indicator table for sediment-associated ecological condition in the sampled Korean freshwater systems. The table summarizes how selected taxa were distributed along the measured PC1 gradient, but it does not imply that KBSI fully isolates sediment effects from water-column conditions or habitat context. This cautious interpretation is consistent with macroinvertebrate bioassessment frameworks, which recognize that benthic taxa integrate multiple stressors and habitat-scale processes rather than responding to a single environmental factor alone (Birk et al., 2012; Hering et al., 2004; Rosenberg and Resh, 1993). The use of taxon-specific tolerance values requires standardized identification at the lowest practicable taxonomic level and may limit routine application where specialist taxonomic expertise is unavailable. Future operational evaluation should determine whether selected taxa can be aggregated to higher taxonomic levels without materially reducing the discriminatory performance of KBSI.

3.4. Relationship between KBSI and sediment environmental variables

Using site-level annual mean values for the 80 monitoring sites, KBSI showed consistent negative relationships with the five sediment environmental variables used to define the PC1 sediment-condition gradient (Fig. 4). KBSI decreased with increasing porewater TAN, AVS, TOC, Metals Tier II Index, and mud fraction, and all relationships were statistically significant. At the site level, the correlation coefficients were -0.57 for TAN, -0.60 for AVS, -0.65 for TOC, -0.64 for the Metals Tier II Index, and -0.51 for mud fraction. Thus, the strongest associations were observed for TOC and Metals Tier II Index, followed by AVS, TAN, and mud fraction. These results indicate that lower KBSI values were associated with

depositional sediments characterized by greater organic matter content, greater relative sediment-bound metal contamination, stronger reducing conditions, higher porewater nitrogen enrichment, and greater fine-particle accumulation.

The negative relationship between KBSI and TOC suggests that the index responded to organic matter accumulation in surface sediments. Organic matter enrichment can increase microbial oxygen demand and promote low-oxygen or reducing conditions in benthic habitats, thereby affecting sediment-dwelling fauna and benthic community composition (Eggleton and Thomas, 2004; Rosenberg et al., 2001). TOC can also interact with redox condition and contaminant binding, so its relationship with KBSI should be interpreted as part of a broader sediment biogeochemical gradient rather than as an isolated single-stressor effect (Zhang et al., 2014).

KBSI also decreased with increasing TAN. In this study, TAN was interpreted as a porewater indicator of nitrogenous organic enrichment rather than as a stand-alone toxicant gradient. Ammonia and other inorganic nitrogen forms can affect aquatic organisms, but their effects depend on chemical speciation, pH, temperature, oxygen condition, and exchange between porewater and overlying water (Camargo and Alonso, 2006; U.S. EPA, 2013). Therefore, the KBSI–TAN relationship is best interpreted as a field-level association with nitrogenous enrichment and porewater condition within depositional sediments.

The negative relationship between KBSI and AVS indicates that lower index values were associated with more reduced sediment conditions. AVS is important in sediment assessment because sulfide-rich conditions reflect reducing environments and can modify both porewater chemistry and metal partitioning (Di Toro et al., 1992). However, AVS should not be interpreted simply as a direct stressor with a single directional effect, because sulfides can bind divalent metals and reduce their bioavailability under some geochemical conditions (Di

Toro et al., 1992; Zhang et al., 2014). Accordingly, the KBSI–AVS relationship should be interpreted as a biological response associated with reduced, organic-rich depositional conditions rather than as direct evidence of sulfide toxicity.

KBSI decreased with increasing Metals Tier II Index, suggesting that assemblages with lower KBSI values tended to occur where relative sediment-bound metal contamination was higher. Sediments can act as both sinks and secondary sources of metals, and metal mobility and bioavailability are influenced by particle size, organic matter, Fe–Mn oxides, AVS, pH, redox condition, and biological activity (Eggleton and Thomas, 2004; Förstner and Wittmann, 1981). Therefore, the relationship between KBSI and the Metals Tier II Index should be interpreted as a field-level association between benthic assemblage condition and relative sediment-bound metal contamination, not as a direct measurement of metal toxicity. This distinction is important because total or guideline-normalized metal concentrations do not necessarily represent the fraction that is bioavailable to benthic organisms (Di Toro et al., 1992; Zhang et al., 2014).

KBSI also showed a statistically significant but moderate negative relationship with mud fraction, indicating a general tendency for lower KBSI values to occur in depositional sediments with a greater silt–clay proportion. However, mud fraction alone accounted for only a limited proportion of the variation in KBSI and should not be interpreted as the dominant determinant of index values. Fine sediments can alter benthic habitats by reducing interstitial flow, filling substrate spaces, changing oxygen exchange, and increasing the retention of organic matter and particle-reactive contaminants (Von Bertrab et al., 2013; Wood and Armitage, 1997). Fine-sediment deposition can also influence macroinvertebrate assemblage composition and interact with other stressors such as organic enrichment and water-quality degradation (Doretto et al., 2018; Extence et al., 2010). Nevertheless, mud fraction is not a

chemical contaminant. It can reflect natural depositional and hydromorphological processes as well as anthropogenic changes in flow regulation, sediment supply, land use, or channel modification (Wharton et al., 2023; Wood and Armitage, 1997).

Overall, these relationships support the internal consistency of KBSI with the sediment variables used to define the PC1 gradient. They indicate that KBSI varied consistently along field gradients involving organic enrichment, reducing conditions, relative sediment-bound metal contamination, and fine-particle deposition. However, these associations do not demonstrate that KBSI isolates sediment effects from co-varying water-column conditions or habitat characteristics (Birk et al., 2012; Feld and Hering, 2007; Townsend et al., 2008).

3.5. Relationship between PC1 score and KBSI

Based on site-level annual mean values for the 80 monitoring sites, KBSI showed a strong negative relationship with the PCA-derived PC1 sediment-condition gradient (Fig. 5).

Because PC1 was derived from mutually correlated sediment variables, including TAN, AVS, TOC, Metals Tier II Index, and mud fraction, it summarized the dominant field gradient represented by the measured sediment variables. Ordination-based approaches are commonly used in aquatic ecology to extract synthetic environmental gradients and relate them to biological assemblage patterns (ter Braak and Verdonschot, 1995). Similar gradient-based logic has also been applied in macroinvertebrate index development, where PCA of abiotic variables was used to define a disturbance gradient against which biological condition was evaluated (Blocksom and Johnson, 2009).

KBSI decreased as PC1 scores increased. Under the sign convention used in this study, PC1 scores represented stronger sediment-associated environmental stress, including

nitrogenous enrichment, reducing conditions, organic matter accumulation, relative sediment-bound metal contamination, and mud deposition. The negative PC1–KBSI relationship therefore indicates that the final community-level index retained the intended directionality of the PC1 gradient after taxon-specific tolerance values and indicator weights were integrated across assemblages.

The PC1–KBSI relationship remained significant after excluding four sites outside the 95% prediction interval in the site-level annual mean analysis. This trimming step was not used to redefine KBSI or to exclude sites from the index framework. Rather, it was used to examine whether the PC1–KBSI relationship was driven by a small number of extreme observations. After trimming, the remaining 76 sites still showed a clear negative relationship, indicating that the association between KBSI and PC1 was not dependent only on outliers. This interpretation is consistent with ecological assessment studies that emphasize the need to consider metric uncertainty and site-specific variability when detecting biological responses along environmental gradients (Johnson et al., 2006).

The PC1–KBSI relationship complements the relationships between KBSI and individual sediment variables described in Section 3.4. Individual variables describe specific sediment attributes, whereas PC1 summarizes their combined field covariation. PCA-based composite approaches can reduce correlated environmental variables into interpretable summary gradients; however, such gradients should be interpreted as integrated field conditions rather than as direct causal mechanisms (Praus, 2019). Therefore, the PC1–KBSI relationship supports the internal coherence of KBSI with the sediment-condition gradient used for its development. However, this result should not be interpreted as external validation of KBSI. Because PC1 was also used to derive the taxon-specific tolerance values and indicator weights, the relationship between PC1 and KBSI represents an internal consistency check

rather than an independent validation test.

3.6. Observed–calculated KBSI and internal agreement

Observed KBSI values showed a positive relationship with calculated KBSI values (Fig. 6). After excluding four sites that fell outside the 95% prediction interval in the preceding site-level PC1–KBSI analysis, the remaining 76 sites retained a clear observed–calculated relationship. This result indicates that the fitted sediment-variable model reproduced the overall variation in site-level KBSI within the development dataset.

Because the calculated values were derived within the same sediment-variable and index-development framework, the observed–calculated relationship should be interpreted as an internal agreement check rather than as evidence of predictive performance or external validation. Nevertheless, the agreement supports the internal coherence of KBSI as a biological indicator associated with the field-derived sediment-condition gradient. KBSI should therefore be interpreted together with sediment chemistry and habitat information, rather than as a stand-alone measure of sediment contamination or toxicity.

3.7. Limitations and future validation

Water-column variables, including dissolved oxygen, conductivity, nutrients, and temperature, were not measured consistently and concurrently across all site-survey sample units. Standardized hydromorphological and habitat variables were also unavailable for the complete dataset. Consequently, the independent and shared contributions of sediment condition, water-column quality, and habitat setting could not be separated statistically. Because macroinvertebrate assemblages may respond non-specifically to interacting

environmental stressors, the observed biological response along PC1 does not by itself isolate causal sediment effects from co-varying water-column or habitat conditions (Heß et al., 2024). KBSI should therefore be interpreted as an integrated biological response associated with sediment conditions under field settings rather than as a sediment-specific measure of toxicity. Future studies should incorporate co-located water-column and hydromorphological measurements to quantify their relative contributions.

The sampling framework was specifically designed for depositional sediment habitats in which fine particles and associated organic or reduced materials accumulate. Accordingly, the applicability of KBSI to coarse riffle, gravel-, or boulder-dominated habitats was not evaluated and should not be assumed. Biological samples were collected to a depth of 20 cm to include deeper-burrowing taxa, whereas sediment physicochemical variables were measured in the upper 0–7 cm layer representing the biologically active surface sediment. Further validation is needed to determine whether KBSI responses remain consistent across contrasting depositional environments and waterbody types.

The same monitoring dataset was used to derive taxon-specific tolerance values and indicator weights and to evaluate the relationships of KBSI with sediment variables and the PC1 gradient. Consequently, the reported relationships represent internal consistency within the index-development dataset rather than independent validation, and their apparent strength may be optimistic. No geographically or temporally independent dataset collected using the same sampling and taxonomic protocols was available for external validation. Seasonal effects were also not evaluated separately because the repeated surveys were not designed as a balanced seasonal experiment. Independent validation using additional sites, temporally separate surveys, and seasonally balanced monitoring data is therefore needed to evaluate the predictive accuracy, temporal stability, and transferability of KBSI.

Overall, KBSI provides a field-based biological indicator of sediment-associated ecological condition in Korean freshwater systems by integrating taxon-specific tolerance values, indicator weights, and community composition along a field-derived sediment gradient. Nevertheless, it should be interpreted together with sediment chemistry and habitat information rather than as a stand-alone measure of sediment contamination or toxicity.

4. Conclusions

This study developed the Korean Benthic Macroinvertebrate Sediment Index (KBSI) as a field-based biological indicator of sediment-associated ecological condition in Korean freshwater systems. The index was derived from a PCA-based sediment gradient integrating porewater TAN, AVS, TOC, the Metals Tier II Index, and mud fraction. Occurrence patterns of 137 indicator taxa across nine PC1 class intervals were used to derive taxon-specific tolerance values and indicator weights.

KBSI decreased consistently with increasing TAN, AVS, TOC, relative sediment-bound metal contamination, mud fraction, and the integrated PC1 gradient. These relationships demonstrate that KBSI retained the intended biological directionality of the sediment-condition gradient within the development dataset. The observed–calculated relationship further supported internal agreement of the index formulation, although it did not constitute independent validation.

KBSI is intended for biological assessment of depositional sediment habitats and should not be interpreted as an independent chemical sediment-quality standard or direct toxicological endpoint. Its results should be considered together with sediment physicochemical and habitat information. Independent validation across additional sites,

648 survey periods, and contrasting depositional environments is required to establish its broader

649 robustness and transferability.

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CRedit authorship contribution statement

Jeaha Song: Conceptualization, Methodology, Visualization, Data validation and curation, formal analysis, Writing-original draft preparation, Writing-review and editing. **Chanyoung Jeong:** Conceptualization, Methodology, Writing-review and editing. **Ihn-Sil Kwak:** Writing-review and editing. **Dongsoo Kong:** Conceptualization, Methodology, Writing-review and editing, Project administration.

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**Development of the Korean Benthic Macroinvertebrate Sediment Index
(KBSI) for assessing sediment-associated ecological condition in freshwater
ecosystems**

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Abstract

Freshwater sediments accumulate organic matter, nutrients, metals, and fine particles and can influence benthic macroinvertebrate assemblages. However, biological tools for evaluating sediment-associated ecological condition remain limited in Korean freshwater systems. This study developed the Korean Benthic Macroinvertebrate Sediment Index (KBSI) as a field-based biological indicator using data from 80 sites and 320 site-survey sample units. Sediment variables, including porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), the Metals Tier II Index, and mud fraction, were summarized using principal component analysis. PC1 represented an integrated sediment-condition gradient associated with organic enrichment, reducing conditions, relative sediment-bound metal contamination, and fine-particle accumulation. PC1 scores from 309 sample units containing selected indicator taxa were divided into nine class intervals using a Weibull cumulative distribution function and Sturges' rule. Taxon-specific tolerance values and indicator weights were derived from occurrence distributions along this gradient and used to calculate KBSI. KBSI showed significant negative relationships with TAN, AVS, TOC, the Metals Tier II Index, mud fraction, and PC1, indicating that lower index values were associated with more degraded sediment-associated field conditions. The relationship between observed and calculated KBSI values supported internal agreement within the development dataset but did not constitute independent validation. KBSI values were classified into five ecological condition classes from A to E for practical interpretation. KBSI is intended for biological assessment of depositional sediment habitats and should be interpreted together with sediment physicochemical and habitat information rather than as an independent chemical sediment-quality standard or direct toxicological endpoint.

Keywords: Benthic macroinvertebrates; Biological assessment; Freshwater sediment; KBSI;

40 Principal component analysis; Sediment-associated ecological condition

41

1. Introduction

Sediment in rivers and lakes is a particulate matrix of gravel, sand, silt, clay, and organic matter that stores watershed-derived pollutants over extended periods and exchanges solutes with the water column as a dynamic reactive system (Chapman, 1990; Zhou et al., 2024). Settling and resuspension, sorption and desorption, and redox-mediated transformations continually modify sediment properties (El-Sharkawy et al., 2025; Hiki et al., 2023). Where fine fractions dominate, contaminant accumulation and bioavailability increase and can restructure benthic macroinvertebrate assemblages (Relyea et al., 2012; Wharton et al., 2023). Therefore, single-time measurements of water-column concentrations do not adequately represent cumulative contamination or habitat alteration, underscoring the need for biological indicators that reflect sediment-associated ecological conditions.

The Oligochaete Index of Sediment Bioindication (IOBS) uses oligochaete assemblage composition to evaluate sediment-associated biological condition in relation to sediment impairment (AFNOR, 2002; AFNOR, 2016), and the U.S. Equilibrium Partitioning Sediment Benchmarks (ESB) quantify linkages between sediment chemistry and biological effects (U.S.EPA, 2003). The Sediment Quality Triad (SQT) integrates chemistry, toxicity testing, and community data to diagnose ecological degradation in contaminated sediments (Chapman, 1990). Stress-specific biotic indices have also been developed: the Fine Sediment Biotic Index (FSBI) encodes sensitivities to fine particles (Relyea et al., 2012), and the Proportion of Sediment-sensitive Invertebrates (PSI) uses the share of sensitive taxa to evaluate restoration outcomes (Extence et al., 2010).

In coastal and estuarine environments, the Benthic Response Index (BRI) and the AZTI Marine Biotic Index (AMBI) apply weighted-tolerance principles and have informed analogous approaches for fresh waters (Borja et al., 2000; Smith et al., 2001). More recently,

multivariate formulations—such as the Hellenic Lake Littoral Biotic Index (HeLLBI) and the Cumulative Biotic Index (CBI)—have related community structure to pollution gradients within explicit statistical frameworks, while high-throughput metabarcoding is improving congruence with morphology-based indices (Lobo et al., 2017; Mavromati et al., 2021; Sudarso et al., 2026).

In Korea, nationally standardized tools for water-column conditions—such as the Korean Benthic macroinvertebrate Index of Biotic Integrity (KB-IBI)—are established, but sediment-focused indicators remain comparatively underdeveloped (Jun et al., 2012). A domestic prototype, the Benthic macroinvertebrate Sediment Index (BSI), estimated taxon tolerance values and indicator weights from relative frequencies in freshwater sediments and computed a weighted mean using a Sediment Quality Index that integrates total organic carbon (TOC), ammonia ($\text{NH}_3\text{-N}$), and heavy metals (Kim et al., 2019). Although BSI showed statistical associations with sediment impairment gradients, its generality was constrained by limited sample size and taxonomic uncertainty within Chironomidae, indicating the need for a tool calibrated with broader spatial coverage and refined taxonomy.

To address this gap, we developed the Korean Benthic Macroinvertebrate Sediment Index (KBSI) as a field-based biological indicator of sediment-associated ecological condition in Korean freshwaters. Specifically, this study aimed to: (1) assemble a nationwide dataset of co-located benthic macroinvertebrate assemblages and sediment physicochemical conditions from major Korean river basins and representative waterbody types; (2) define an integrated sediment environmental gradient using PCA of porewater total ammonia nitrogen, acid-volatile sulfides, total organic carbon, heavy metals, and the mud fraction; (3) derive taxon-specific tolerance values and indicator weights from macroinvertebrate occurrence patterns along the PC1-based sediment gradient; and (4) formulate and examine KBSI as a community-level

index by relating index values to individual sediment variables, the PC1 sediment gradient, and observed–calculated index agreement. KBSI is therefore intended not as a stand-alone chemical or toxicological measure of sediment quality, but as a biological indicator of benthic assemblage responses to sediment-associated environmental conditions under field settings.

2. Materials and methods

2.1 Study area and sampling design

2.1.1 Site description

This study was conducted at 80 freshwater sediment monitoring sites distributed across the major river basins of Korea (Fig. 1). The sites were selected to represent broad gradients in sedimentary environments, waterbody type, and anthropogenic influence, including both lotic and lentic systems. For descriptive and management-oriented interpretation, the sites were categorized into five groups: general streams, urban streams, industrial streams, lakes, and reservoirs. The stream categories were assigned according to the dominant surrounding land-use setting and the original monitoring-site designation, rather than using uniform quantitative catchment-scale land-use thresholds. Urban streams represented lotic sites located in predominantly urbanized settings, whereas industrial streams represented sites associated with industrial areas. General streams comprised the remaining lotic sites that were not classified as urban or industrial streams.

These categories were used to describe broad differences in waterbody type, surrounding land-use setting, and sediment depositional environment, rather than to define strict reference or impairment classes. General streams were therefore not treated as reference or best-available sites and could include streams influenced by agriculture, forestry, flow

regulation, impoundments, or other catchment and hydromorphological pressures.

Field surveys were conducted repeatedly between 2022 and 2024, producing a total of 320 site-survey sample units. For the derivation of taxon-specific tolerance values and indicator weights, 309 sample units in which selected indicator taxa occurred were used. Sample units without selected indicator taxa were excluded from the tolerance-value calculation because they did not provide the taxon-occurrence information required to estimate biological responses along the sediment environmental gradient.

The sampling design was developed to support a field-based biological assessment of sediment-associated ecological condition. Accordingly, biological and sediment samples were collected from comparable depositional habitats within each site, where fine sediments and associated organic or reduced materials could accumulate. The sampling framework was therefore intended for depositional sediment habitats, not for coarse riffle or boulder-dominated substrates. This design allowed the KBSI to characterize benthic macroinvertebrate assemblage responses to sediment-associated environmental conditions under field settings.

2.1.2. Collection and Identification of Benthic Macroinvertebrates

Benthic macroinvertebrates were collected from depositional sediment habitats using a cylindrical core sampler with an internal diameter of 7.5 cm. At each site, five replicate cores were taken to a depth of 20 cm where coring was feasible (Kim et al., 2018; Song et al., 2025). The 20 cm sampling depth was adopted to capture not only organisms concentrated in the upper sediment layer, but also burrowing taxa, particularly oligochaetes, that may occur in deeper sediment layers.

In the field, sediment core contents were washed through a 0.20-mm sieve to minimize

the loss of small-bodied sediment-dwelling organisms, particularly oligochaetes and chironomid larvae. Retained materials were preserved in ethanol and transported to the laboratory. The use of a 0.20-mm sieve was selected because small-bodied taxa in fine sediments can be underestimated when larger mesh sizes are used (Song et al., 2025).

In the laboratory, residual sediment was removed and organisms were sorted under a stereomicroscope. Specimens were identified to the lowest practicable taxonomic level, generally to species where diagnostic characters were available, using standard taxonomic references (Cummins and Merritt, 1996; Kawai and Tanida, 2005; Peckarsky et al., 1990; Yoon, 1988). Chironomid larvae were identified based on diagnostic structures such as the head capsule, mentum, mandible, and antennae (Kwak, 2015). Oligochaetes were identified using external and reproductive morphological characters, including chaetal and reproductive structures (Park et al., 2016). All identified specimens were preserved in 80% ethanol to minimize morphological deformation.

Taxa that were sufficiently frequent and taxonomically reliable were retained as candidate indicator taxa for KBSI development. This taxonomic screening was necessary because the derivation of tolerance values and indicator weights requires reliable occurrence information along the sediment environmental gradient.

2.1.3. Sediment Sampling and Physicochemical Analysis

Surface sediment samples for physicochemical analysis were collected concurrently with the biological samples from the corresponding depositional habitats within each site using a cylindrical core sampler with an internal diameter of 7.5 cm. For sediment chemistry and particle-size analysis, the upper 0–7 cm sediment layer was used because this layer represents

the biologically active surface sediment where organic matter accumulation, redox gradients, porewater chemistry, and contaminant binding are most pronounced. This analytical depth differed from the macroinvertebrate sampling depth of 20 cm, which was used to include deeper-burrowing taxa.

Each sediment sample was sealed in a 300-mL high-density polyethylene bottle, stored at ≤ 4 °C, and transported to the laboratory. The target sediment variables were porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), heavy metals, and particle-size composition. TAN was used as an indicator of nitrogenous organic enrichment in sediment porewater, AVS represented reduced sediment conditions associated with sulfide formation, and TOC represented organic matter accumulation.

TAN was quantified colorimetrically at 630 nm. TOC was measured using a CHNS analyzer after removal of inorganic carbon. AVS was determined as Ag₂S precipitated under a nitrogen stream, followed by filtration and gravimetric drying. For heavy metals, sediment samples were digested with mixed acids and analyzed using ICP–AES or ICP–MS; Hg was quantified using TDA–AAS. Relative sediment-bound metal contamination was summarized using the Metals Tier II Index based on the Tier II sediment guideline values for Cu, Pb, Ni, As, Hg, Zn, Cd, and Cr (NIER, 2022).

$$\text{Metals Tier II Index} = \frac{\sum_{i=1}^8 \left(\frac{EC_i}{PELK_i} \right)}{8} \quad (1)$$

Where EC_i is the measured concentration of metal i and $PELK_i$ is the corresponding Tier II guideline value. Higher values indicate greater relative sediment-bound metal contamination compared with the Tier II sediment guideline benchmarks.

Particle-size fractions were separated using 2-mm and 75- μ m sieves into gravel (>2 mm), sand (0.075–2 mm), and silt–clay (<0.075 mm) fractions. In this study, the fine-particle

variable used for KBSI development refers to the silt–clay fraction, hereafter termed the mud fraction, and does not include sand. The mud fraction was treated as a physical depositional variable rather than a chemical contaminant, because fine particles can co-vary with organic matter accumulation, reduced conditions, and contaminant retention in depositional habitats. Routine quality assurance and quality control procedures followed the national sediment monitoring guidelines (NIER, 2022).

2.2. Development and calculation of the Korean Benthic Macroinvertebrate Sediment Index (KBSI)

2.2.1. PCA-derived sediment environmental gradient

To derive an integrated sediment environmental gradient for KBSI development, principal component analysis (PCA) was performed using the sediment physicochemical variables measured at each sample unit. PCA was used to reduce redundancy among correlated environmental variables and to summarize the dominant multivariate gradient in sediment condition (Abdi and Williams, 2010; Jolliffe and Cadima, 2016). The variables included porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), the Metals Tier II Index, and the mud fraction. These variables were selected to represent major sediment-associated conditions, including nitrogenous organic enrichment, reduced sediment conditions, organic matter accumulation, relative sediment-bound metal contamination, and fine-particle deposition.

Prior to PCA, variables with skewed distributions were transformed to reduce the influence of extreme values. TAN, AVS, TOC, and the Metals Tier II Index were log-transformed, whereas the mud fraction, expressed as proportional data, was arcsine square-root

transformed. All transformed variables were standardized to a mean of zero and a standard deviation of one before analysis, and PCA was conducted on the standardized environmental matrix (Legendre and Legendre, 2012; Quinn and Keough, 2002).

The first principal component score was used as the integrated sediment environmental gradient for deriving taxon-specific tolerance values and indicator weights. PC1 was selected because it represented the dominant axis of variation in sediment conditions and provided a continuous gradient along which macroinvertebrate occurrence patterns could be evaluated. In this study, PC1 was not interpreted as a stand-alone chemical pollution index or an independent measure of sediment toxicity. Instead, it was used as a multivariate sediment-condition gradient reflecting field-based co-variation among sediment physicochemical variables.

The PC1 scores of sample units containing selected indicator taxa were used in the subsequent derivation of biological indicator values. Among the 320 site-survey sample units, 309 sample units with selected indicator taxa were retained for this step, whereas sample units without selected indicator taxa were excluded because they did not provide occurrence information required for estimating taxon responses along the sediment environmental gradient.

2.2.2. Classification of the PC1 gradient and occurrence distribution

To classify the PC1-based sediment environmental gradient for deriving biological indicator values, the distribution of PC1 scores was analyzed using sample units in which selected indicator taxa occurred. Among the 320 site-survey sample units, 309 sample units were retained for this step after excluding 11 sample units in which no selected indicator taxa were recorded.

The distribution of PC1 scores for the 309 sample units was fitted using a Weibull

cumulative distribution function, which was selected because of its flexibility in describing skewed environmental-gradient distributions and deriving probability-based class boundaries (Weibull, 1951). The Weibull cumulative distribution function was expressed as:

$$F(x) = 1 - \exp[-(\frac{x-\theta}{\lambda})^k] \quad (2)$$

where $F(x)$ is the cumulative probability at PC1 score x , θ is the location parameter, λ is the scale parameter, and k is the shape parameter.

The number of PC1 class intervals was determined using Sturges' rule, resulting in nine class intervals (Sturges, 1926):

$$K = 1 + \log_2(n) \quad (3)$$

where K is the number of class intervals and n is the number of sample units. For the 309 sample units used in this step, Sturges' rule produced approximately nine intervals.

Class boundaries were obtained from the inverse Weibull cumulative distribution function so that each PC1 interval represented an equal probability range:

$$x_p = \theta + \lambda[-\ln(1 - p)]^{1/k} \quad (4)$$

where x_p is the PC1 score corresponding to cumulative probability p . Each interval was assigned a representative class value, which was subsequently used to summarize the occurrence distribution of each indicator taxon along the PC1 gradient. This procedure provided a standardized basis for estimating taxon-specific tolerance values and indicator weights from field occurrence data.

In this framework, the PC1 classes were not treated as regulatory sediment-quality standards. Rather, they were used as empirical intervals of the multivariate sediment-condition gradient to quantify biological occurrence patterns. This interpretation is important because

PC1 includes both chemical variables and a physical depositional variable, the mud fraction, which may reflect natural hydromorphological processes as well as anthropogenic sediment accumulation.

2.2.3. Derivation of taxon-specific tolerance values and indicator weights

Taxon-specific tolerance values and indicator weights were derived from the occurrence distribution of selected indicator taxa along the PC1-based sediment environmental gradient. Only taxa with reliable taxonomic resolution and sufficient occurrence frequency were retained as indicator taxa, because rare or taxonomically uncertain taxa may produce unstable tolerance estimates. This screening was intended to ensure that KBSI was based on taxa with interpretable field occurrence patterns along the sediment-condition gradient. This occurrence-based approach follows the general logic of deriving biological indicator values from taxon distributions along environmental gradients (Kim et al., 2019; Kong and Kim, 2016). The final indicator set therefore represents a screened subset of taxa suitable for KBSI calibration rather than a complete inventory of all taxa collected during the surveys.

For each indicator taxon, its occurrence frequency was calculated within each of the nine PC1 class intervals. The occurrence frequency of taxon i in class k was expressed as:

$$RF_{ik} = \frac{n_{ik}}{N_k} \times 100 \quad (5)$$

where RF_{ik} is the relative occurrence frequency of taxon i in PC1 class k , n_{ik} is the number of sample units in class k where taxon i occurred, and N_k is the total number of sample units in class k .

The occurrence frequencies of each taxon across the nine PC1 classes were then

normalized to obtain its occurrence valency:

$$V_{ik} = \frac{RF_{ik}}{\sum_{k=1}^9 RF_{ik}} \quad (6)$$

where V_{ik} represents the proportional occurrence valency of taxon i in class k . Taxon-specific tolerance values were calculated as the weighted average of the PC1 class values using occurrence valencies as weights:

$$t_i = \sum_{k=1}^9 C_k V_{ik} \quad (7)$$

where t_i is the tolerance value of taxon i , and C_k is the representative class value assigned to PC1 class k . Higher tolerance values indicate that a taxon occurred more frequently under sediment conditions associated with higher PC1-derived stress levels, whereas lower values indicate occurrence concentrated under less degraded sediment conditions.

Indicator weights were assigned according to the concentration of each taxon's normalized occurrence valencies across the nine PC1 classes. Taxa whose occurrence valencies were concentrated within a limited portion of the PC1 gradient were assigned higher indicator weights, whereas taxa with occurrence valencies distributed broadly across multiple classes were assigned lower weights. Thus, the indicator weight represents the specificity of a taxon's occurrence distribution along the sediment-condition gradient, rather than its tolerance position or physiological sensitivity.

This distinction between tolerance value and indicator weight prevents taxa with similar tolerance positions but different occurrence breadths from being treated as equivalent. In the final indicator table, taxa with strongly concentrated occurrence distributions were assigned higher indicator weights, whereas taxa occurring across broader portions of the PC1 gradient were assigned lower weights.

2.2.4. Calculation and classification of the Korean Benthic Macroinvertebrate Sediment Index (KBSI)

The Korean Benthic Macroinvertebrate Sediment Index (KBSI) was calculated as a weighted community-level index using the taxon-specific tolerance value, frequency value, and indicator weight of each indicator taxon. For each sample unit, taxa included in the final indicator list were matched with their corresponding tolerance values and indicator weights, and their frequency values were used to represent their relative contribution to the assemblage. This weighted-index structure follows the general approach of combining taxon tolerance, occurrence, and indicator strength in benthic macroinvertebrate bioassessment (Kim et al., 2019).

The KBSI was calculated as:

$$KBSI = (10 - \frac{\sum_{i=1}^n t_i h_i g_i}{\sum_{i=1}^n h_i g_i}) \times 10 \quad (8)$$

where n is the number of indicator taxa occurring in a sample unit, t_i is the tolerance value of taxon i , h_i is the frequency value of taxon i , and g_i is the indicator weight of taxon i .

The weighted mean tolerance value was subtracted from 10 so that higher KBSI values indicate better sediment-associated ecological condition, whereas lower values indicate assemblages dominated by taxa tolerant of degraded sediment conditions.

KBSI values were scaled from 0 to 100 and classified into five ecological condition classes. For practical interpretation, KBSI values were classified into five assessment classes following the KBSI assessment framework developed in this study: A, high, 70–100; B, good,

60–<70; C, moderate, 50–<60; D, poor, 40–<50; and E, bad, 0–<40.

These classes provide interpretive categories for biological assessment of sediment-associated ecological condition. They should not be interpreted as independent chemical sediment-quality standards, regulatory criteria, or toxicological thresholds. Rather, they summarize KBSI values within the assessment framework developed for this study.

3. Results and discussion

3.1. PCA-derived sediment environmental gradient

Principal component analysis summarized the covariance structure of the five sediment physicochemical variables used to characterize sediment-associated field conditions: porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), Metals Tier II Index, and mud fraction. The first three PCA axes explained 86.0% of the total variance, with PC1 accounting for 58.0%, PC2 for 17.2%, and PC3 for 10.8% (Table 1; Fig. 2). PC1 was the dominant axis because all five sediment variables showed clear associations with this axis. The loadings on PC1 were 0.66 for ln(TAN), 0.69 for ln(AVS), 0.86 for ln(TOC), 0.82 for ln(Metals Tier II Index), and 0.76 for ln(mud fraction). Thus, PC1 represented a composite sediment environmental gradient along which nitrogenous enrichment, reducing sediment conditions, organic matter accumulation, relative sediment-bound metal contamination, and fine-particle deposition co-varied.

Organic matter accumulation is ecologically relevant because it can increase oxygen demand in surface sediments and promote conditions unfavorable to oxygen-sensitive benthic fauna (Rosenberg et al., 2001). AVS reflects reduced sediment conditions, and sulfide-rich or strongly reduced sediments can modify metal binding, porewater chemistry, and exposure

conditions for sediment-dwelling organisms (Simpson et al., 2013). TAN was interpreted as a porewater indicator of nitrogenous organic enrichment rather than as a stand-alone toxicant gradient, because ammonia effects in sediments depend on pH, redox condition, and porewater–overlying water exchange. The Metals Tier II Index summarized the relative degree of sediment-bound metal contamination, but its ecological interpretation should consider bioavailability and site-specific geochemical conditions rather than total concentration alone (Simpson et al., 2013). Therefore, the strong contribution of TOC, AVS, TAN, and the Metals Tier II Index indicates that PC1 captured a biogeochemical gradient associated with organic enrichment, reduced sediment conditions, and relative sediment-bound metal contamination.

Mud fraction also contributed strongly to PC1, but its interpretation differs from that of chemical contaminants. Fine particles can reduce interstitial flow, alter oxygen exchange, and modify benthic habitat structure in lotic environments (Wood and Armitage, 1997). Fine-grained sediment can also affect macroinvertebrate assemblages by filling substrate interstices and reducing the availability of coarse-substrate microhabitats used by rheophilic or sediment-sensitive taxa (Bylak and Kukuła, 2022). In addition, fine sediments can enhance the retention of organic matter and particle-reactive contaminants because of their large surface area and depositional stability (Wharton et al., 2023). However, mud accumulation may arise from natural hydromorphological and depositional processes as well as from anthropogenic changes in flow regulation, land use, and sediment supply (Wharton et al., 2023; Wood and Armitage, 1997). Accordingly, mud fraction was treated as a physical depositional variable rather than as a chemical contaminant.

For these reasons, PC1 was interpreted as a field-based multivariate sediment-condition gradient, not as a stand-alone sediment toxicity index or a regulatory sediment-quality standard. Sediment chemistry and benthic community composition can provide

complementary lines of evidence for sediment-associated ecological assessment, but neither line of evidence alone fully resolves causal mechanisms under field conditions (Long and Chapman, 1985). The sediment quality triad framework similarly emphasizes that sediment chemistry, toxicity, and benthic community responses should be interpreted together rather than treated as interchangeable measures (Chapman, 1990). In the present study, KBSI was derived from macroinvertebrate occurrence patterns along the PC1 gradient; therefore, it should be interpreted as a biological indicator of sediment-associated ecological condition under field settings, not as an independent chemical or toxicological assessment of sediment quality.

3.2. Classification of PC1 scores and derivation of class values

Among the 320 site-survey sample units, 309 sample units contained selected indicator taxa and were therefore used to classify the PC1 sediment-condition gradient. The remaining 11 sample units were excluded from this step because they did not contain selected indicator taxa and therefore provided no taxon-occurrence information for deriving tolerance values or indicator weights. The PC1 scores of the 309 retained sample units were used to describe the empirical distribution of sediment-associated field conditions represented by the indicator-taxon dataset.

The distribution of PC1 scores was fitted using a Weibull cumulative distribution function (Fig. 3). The Weibull model was used to describe the probability distribution of PC1 scores and to derive class boundaries along the continuous sediment-condition gradient. Because the objective was to construct empirical intervals for biological indicator derivation rather than to define regulatory sediment-quality thresholds, the distribution was divided into equal cumulative-probability intervals. Sturges' rule produced nine class intervals for the 309 sample units, and the inverse Weibull cumulative distribution function was used to estimate the

PC1 boundary and representative class value for each interval (Table 2).

The resulting PC1 classes covered the observed sediment-condition gradient from low PC1 scores to high PC1 scores. Under the sign convention used in this study, higher PC1 scores represented stronger sediment-associated environmental stress, including greater organic enrichment, reducing conditions, relative sediment-bound metal contamination, and mud accumulation. The representative class values therefore provided a standardized numerical basis for estimating taxon-specific occurrence positions along the sediment-condition gradient. Taxa occurring mainly in lower PC1 classes were assigned lower tolerance values, whereas taxa occurring mainly in higher PC1 classes were assigned higher tolerance values.

Importantly, these PC1 classes should not be interpreted as sediment-quality standards or toxicity thresholds. They are empirical intervals of a field-based multivariate sediment-condition gradient derived from the distribution of PC1 scores in sample units containing indicator taxa. This distinction is necessary because PC1 integrates chemical variables with mud fraction, a physical depositional variable that can reflect both natural hydromorphological processes and anthropogenic sediment accumulation. The class values were therefore used only as an internal scaling framework for deriving KBSI tolerance values and indicator weights, not as independent criteria for classifying sediment contamination or toxicity.

3.3. Taxon-specific tolerance values and indicator weights

Taxon-specific tolerance values and indicator weights were derived from the occurrence distributions of selected indicator taxa across the nine PC1 class intervals (Table 3). In total, 137 indicator taxa were retained in the final indicator table. This approach follows the logic of field-based macroinvertebrate bioassessment, in which taxon scores are inferred from

observed distributions along environmental gradients rather than from a single chemical variable or laboratory endpoint (Hering et al., 2004; Rosenberg and Resh, 1993). It also builds on the saprobic valency concept and its later adaptations, where taxa are characterized by both their position along an environmental gradient and their strength of indication within that gradient (Sládeček, 1967; Yoon et al., 1992; Zelinka and Marvan, 1961).

Although KBSI shares a weighted-indicator structure with the Korean BMI (Kong et al., 2018) and conventional saprobic indices, the indices differ in their target environmental gradients and application domains. BMI is used primarily to assess the general ecological condition of Korean streams, whereas KBSI was derived from depositional sediment habitats in both lotic and lentic systems. Similarly, conventional saprobic values primarily represent taxon distributions along gradients of organic pollution and oxygen conditions, whereas KBSI tolerance values were independently derived along a multivariate sediment-condition gradient integrating TAN, AVS, TOC, relative metal contamination, and mud fraction. Correspondence among taxon classifications may therefore reflect shared responses to organic enrichment and oxygen depletion, but KBSI tolerance values should not be interpreted as conventional saprobic values.

Tolerance values represent the weighted average position of each taxon along the PC1 sediment-condition gradient. Lower values indicate taxa whose occurrences were concentrated in lower PC1 classes, whereas higher values indicate taxa whose occurrences were concentrated in higher PC1 classes. Under the sign convention used in this study, higher PC1 scores represented stronger sediment-associated environmental stress, including greater organic enrichment, reducing conditions, metal contamination, and mud accumulation. Accordingly, tolerance values should be interpreted as field-derived occurrence positions along the measured PC1 gradient, not as fixed physiological toxicity thresholds.

The resulting values showed broad taxon-level variation. Taxa with low tolerance values included *Rhyacodrilus coccineus* ($t = 0.0$), *Chaetogaster limnaei* ($t = 0.0$), *Limnodrilus claparedeianus* ($t = 0.7$), and *Chaetogaster diaphanus* ($t = 1.9$), whereas taxa with higher tolerance values included *Limnodrilus udekemianus* ($t = 7.3$) and *Branchiodrilus hortensis* ($t = 7.1$). These differences are consistent with the general observation that benthic macroinvertebrate assemblages vary along gradients of organic enrichment, oxygen depletion, sediment condition, and habitat disturbance (Townsend et al., 2008; Von Bertrab et al., 2013). However, the values should not be over-interpreted as universal taxon traits because field distributions can also reflect habitat structure, flow regime, sampling depth, and regional species pools.

Indicator weights describe a different property from tolerance values. A high indicator weight means that a taxon's occurrences were concentrated within a restricted portion of the PC1 gradient, whereas a low weight indicates a broader distribution across classes. Thus, indicator weight represents occurrence specificity rather than sensitivity or tolerance itself. This distinction follows the weighted-indicator logic of saprobic and Korean benthic macroinvertebrate indices, where community-level scores combine taxon position, abundance or occurrence, and indicator strength (Zelinka & Marvan, 1961; Yoon et al., 1992; Kong & Kim, 2016; Kim et al., 2019).

This distinction also resolves the issue of taxa with tolerance values of 0.0. In the final table, *Rhyacodrilus coccineus* and *Chaetogaster limnaei* had tolerance values of 0.0 and indicator weights of 5, because their occurrences were concentrated in the lowest PC1 class. Their high indicator weights therefore reflect narrow occurrence specificity, not an automatic assignment based on sensitivity. Conversely, taxa with intermediate or high tolerance values may receive low weights if they are broadly distributed across PC1 classes. This treatment

prevents tolerance value and indicator weight from being conflated into a single sensitivity metric.

Overall, Table 3 should be interpreted as a field-derived indicator table for sediment-associated ecological condition in the sampled Korean freshwater systems. The table summarizes how selected taxa were distributed along the measured PC1 gradient, but it does not imply that KBSI fully isolates sediment effects from water-column conditions or habitat context. This cautious interpretation is consistent with macroinvertebrate bioassessment frameworks, which recognize that benthic taxa integrate multiple stressors and habitat-scale processes rather than responding to a single environmental factor alone (Birk et al., 2012; Hering et al., 2004; Rosenberg and Resh, 1993). The use of taxon-specific tolerance values requires standardized identification at the lowest practicable taxonomic level and may limit routine application where specialist taxonomic expertise is unavailable. Future operational evaluation should determine whether selected taxa can be aggregated to higher taxonomic levels without materially reducing the discriminatory performance of KBSI.

3.4. Relationship between KBSI and sediment environmental variables

Using site-level annual mean values for the 80 monitoring sites, KBSI showed consistent negative relationships with the five sediment environmental variables used to define the PC1 sediment-condition gradient (Fig. 4). KBSI decreased with increasing porewater TAN, AVS, TOC, Metals Tier II Index, and mud fraction, and all relationships were statistically significant. At the site level, the correlation coefficients were -0.57 for TAN, -0.60 for AVS, -0.65 for TOC, -0.64 for the Metals Tier II Index, and -0.51 for mud fraction. Thus, the strongest associations were observed for TOC and Metals Tier II Index, followed by AVS, TAN, and mud fraction. These results indicate that lower KBSI values were associated with

depositional sediments characterized by greater organic matter content, greater relative sediment-bound metal contamination, stronger reducing conditions, higher porewater nitrogen enrichment, and greater fine-particle accumulation.

The negative relationship between KBSI and TOC suggests that the index responded to organic matter accumulation in surface sediments. Organic matter enrichment can increase microbial oxygen demand and promote low-oxygen or reducing conditions in benthic habitats, thereby affecting sediment-dwelling fauna and benthic community composition (Eggleton and Thomas, 2004; Rosenberg et al., 2001). TOC can also interact with redox condition and contaminant binding, so its relationship with KBSI should be interpreted as part of a broader sediment biogeochemical gradient rather than as an isolated single-stressor effect (Zhang et al., 2014).

KBSI also decreased with increasing TAN. In this study, TAN was interpreted as a porewater indicator of nitrogenous organic enrichment rather than as a stand-alone toxicant gradient. Ammonia and other inorganic nitrogen forms can affect aquatic organisms, but their effects depend on chemical speciation, pH, temperature, oxygen condition, and exchange between porewater and overlying water (Camargo and Alonso, 2006; U.S. EPA, 2013). Therefore, the KBSI–TAN relationship is best interpreted as a field-level association with nitrogenous enrichment and porewater condition within depositional sediments.

The negative relationship between KBSI and AVS indicates that lower index values were associated with more reduced sediment conditions. AVS is important in sediment assessment because sulfide-rich conditions reflect reducing environments and can modify both porewater chemistry and metal partitioning (Di Toro et al., 1992). However, AVS should not be interpreted simply as a direct stressor with a single directional effect, because sulfides can bind divalent metals and reduce their bioavailability under some geochemical conditions (Di

Toro et al., 1992; Zhang et al., 2014). Accordingly, the KBSI–AVS relationship should be interpreted as a biological response associated with reduced, organic-rich depositional conditions rather than as direct evidence of sulfide toxicity.

KBSI decreased with increasing Metals Tier II Index, suggesting that assemblages with lower KBSI values tended to occur where relative sediment-bound metal contamination was higher. Sediments can act as both sinks and secondary sources of metals, and metal mobility and bioavailability are influenced by particle size, organic matter, Fe–Mn oxides, AVS, pH, redox condition, and biological activity (Eggleton and Thomas, 2004; Förstner and Wittmann, 1981). Therefore, the relationship between KBSI and the Metals Tier II Index should be interpreted as a field-level association between benthic assemblage condition and relative sediment-bound metal contamination, not as a direct measurement of metal toxicity. This distinction is important because total or guideline-normalized metal concentrations do not necessarily represent the fraction that is bioavailable to benthic organisms (Di Toro et al., 1992; Zhang et al., 2014).

KBSI also showed a statistically significant but moderate negative relationship with mud fraction, indicating a general tendency for lower KBSI values to occur in depositional sediments with a greater silt–clay proportion. However, mud fraction alone accounted for only a limited proportion of the variation in KBSI and should not be interpreted as the dominant determinant of index values. Fine sediments can alter benthic habitats by reducing interstitial flow, filling substrate spaces, changing oxygen exchange, and increasing the retention of organic matter and particle-reactive contaminants (Von Bertrab et al., 2013; Wood and Armitage, 1997). Fine-sediment deposition can also influence macroinvertebrate assemblage composition and interact with other stressors such as organic enrichment and water-quality degradation (Doretto et al., 2018; Extence et al., 2010). Nevertheless, mud fraction is not a

chemical contaminant. It can reflect natural depositional and hydromorphological processes as well as anthropogenic changes in flow regulation, sediment supply, land use, or channel modification (Wharton et al., 2023; Wood and Armitage, 1997).

Overall, these relationships support the internal consistency of KBSI with the sediment variables used to define the PC1 gradient. They indicate that KBSI varied consistently along field gradients involving organic enrichment, reducing conditions, relative sediment-bound metal contamination, and fine-particle deposition. However, these associations do not demonstrate that KBSI isolates sediment effects from co-varying water-column conditions or habitat characteristics (Birk et al., 2012; Feld and Hering, 2007; Townsend et al., 2008).

3.5. Relationship between PC1 score and KBSI

Based on site-level annual mean values for the 80 monitoring sites, KBSI showed a strong negative relationship with the PCA-derived PC1 sediment-condition gradient (Fig. 5). Because PC1 was derived from mutually correlated sediment variables, including TAN, AVS, TOC, Metals Tier II Index, and mud fraction, it summarized the dominant field gradient represented by the measured sediment variables. Ordination-based approaches are commonly used in aquatic ecology to extract synthetic environmental gradients and relate them to biological assemblage patterns (ter Braak and Verdonschot, 1995). Similar gradient-based logic has also been applied in macroinvertebrate index development, where PCA of abiotic variables was used to define a disturbance gradient against which biological condition was evaluated (Blocksom and Johnson, 2009).

KBSI decreased as PC1 scores increased. Under the sign convention used in this study, PC1 scores represented stronger sediment-associated environmental stress, including

nitrogenous enrichment, reducing conditions, organic matter accumulation, relative sediment-bound metal contamination, and mud deposition. The negative PC1–KBSI relationship therefore indicates that the final community-level index retained the intended directionality of the PC1 gradient after taxon-specific tolerance values and indicator weights were integrated across assemblages.

The PC1–KBSI relationship remained significant after excluding four sites outside the 95% prediction interval in the site-level annual mean analysis. This trimming step was not used to redefine KBSI or to exclude sites from the index framework. Rather, it was used to examine whether the PC1–KBSI relationship was driven by a small number of extreme observations. After trimming, the remaining 76 sites still showed a clear negative relationship, indicating that the association between KBSI and PC1 was not dependent only on outliers. This interpretation is consistent with ecological assessment studies that emphasize the need to consider metric uncertainty and site-specific variability when detecting biological responses along environmental gradients (Johnson et al., 2006).

The PC1–KBSI relationship complements the relationships between KBSI and individual sediment variables described in Section 3.4. Individual variables describe specific sediment attributes, whereas PC1 summarizes their combined field covariation. PCA-based composite approaches can reduce correlated environmental variables into interpretable summary gradients; however, such gradients should be interpreted as integrated field conditions rather than as direct causal mechanisms (Praus, 2019). Therefore, the PC1–KBSI relationship supports the internal coherence of KBSI with the sediment-condition gradient used for its development. However, this result should not be interpreted as external validation of KBSI. Because PC1 was also used to derive the taxon-specific tolerance values and indicator weights, the relationship between PC1 and KBSI represents an internal consistency check

rather than an independent validation test.

3.6. Observed–calculated KBSI and internal agreement

Observed KBSI values showed a positive relationship with calculated KBSI values (Fig. 6). After excluding four sites that fell outside the 95% prediction interval in the preceding site-level PC1–KBSI analysis, the remaining 76 sites retained a clear observed–calculated relationship. This result indicates that the fitted sediment-variable model reproduced the overall variation in site-level KBSI within the development dataset.

Because the calculated values were derived within the same sediment-variable and index-development framework, the observed–calculated relationship should be interpreted as an internal agreement check rather than as evidence of predictive performance or external validation. Nevertheless, the agreement supports the internal coherence of KBSI as a biological indicator associated with the field-derived sediment-condition gradient. KBSI should therefore be interpreted together with sediment chemistry and habitat information, rather than as a stand-alone measure of sediment contamination or toxicity.

3.7. Limitations and future validation

Water-column variables, including dissolved oxygen, conductivity, nutrients, and temperature, were not measured consistently and concurrently across all site-survey sample units. Standardized hydromorphological and habitat variables were also unavailable for the complete dataset. Consequently, the independent and shared contributions of sediment condition, water-column quality, and habitat setting could not be separated statistically. Because macroinvertebrate assemblages may respond non-specifically to interacting

environmental stressors, the observed biological response along PC1 does not by itself isolate causal sediment effects from co-varying water-column or habitat conditions (Heß et al., 2024). KBSI should therefore be interpreted as an integrated biological response associated with sediment conditions under field settings rather than as a sediment-specific measure of toxicity. Future studies should incorporate co-located water-column and hydromorphological measurements to quantify their relative contributions.

The sampling framework was specifically designed for depositional sediment habitats in which fine particles and associated organic or reduced materials accumulate. Accordingly, the applicability of KBSI to coarse riffle, gravel-, or boulder-dominated habitats was not evaluated and should not be assumed. Biological samples were collected to a depth of 20 cm to include deeper-burrowing taxa, whereas sediment physicochemical variables were measured in the upper 0–7 cm layer representing the biologically active surface sediment. Further validation is needed to determine whether KBSI responses remain consistent across contrasting depositional environments and waterbody types.

The same monitoring dataset was used to derive taxon-specific tolerance values and indicator weights and to evaluate the relationships of KBSI with sediment variables and the PC1 gradient. Consequently, the reported relationships represent internal consistency within the index-development dataset rather than independent validation, and their apparent strength may be optimistic. No geographically or temporally independent dataset collected using the same sampling and taxonomic protocols was available for external validation. Seasonal effects were also not evaluated separately because the repeated surveys were not designed as a balanced seasonal experiment. Independent validation using additional sites, temporally separate surveys, and seasonally balanced monitoring data is therefore needed to evaluate the predictive accuracy, temporal stability, and transferability of KBSI.

Overall, KBSI provides a field-based biological indicator of sediment-associated ecological condition in Korean freshwater systems by integrating taxon-specific tolerance values, indicator weights, and community composition along a field-derived sediment gradient. Nevertheless, it should be interpreted together with sediment chemistry and habitat information rather than as a stand-alone measure of sediment contamination or toxicity.

4. Conclusions

This study developed the Korean Benthic Macroinvertebrate Sediment Index (KBSI) as a field-based biological indicator of sediment-associated ecological condition in Korean freshwater systems. The index was derived from a PCA-based sediment gradient integrating porewater TAN, AVS, TOC, the Metals Tier II Index, and mud fraction. Occurrence patterns of 137 indicator taxa across nine PC1 class intervals were used to derive taxon-specific tolerance values and indicator weights.

KBSI decreased consistently with increasing TAN, AVS, TOC, relative sediment-bound metal contamination, mud fraction, and the integrated PC1 gradient. These relationships demonstrate that KBSI retained the intended biological directionality of the sediment-condition gradient within the development dataset. The observed–calculated relationship further supported internal agreement of the index formulation, although it did not constitute independent validation.

KBSI is intended for biological assessment of depositional sediment habitats and should not be interpreted as an independent chemical sediment-quality standard or direct toxicological endpoint. Its results should be considered together with sediment physicochemical and habitat information. Independent validation across additional sites,

648 survey periods, and contrasting depositional environments is required to establish its broader
649 robustness and transferability.

650

CRedit authorship contribution statement

Jeaha Song: Conceptualization, Methodology, Visualization, Data validation and curation, formal analysis, Writing-original draft preparation, Writing-review and editing. **Chanyoung Jeong:** Conceptualization, Methodology, Writing-review and editing. **Ihn-Sil Kwak:** Writing-review and editing. **Dongsoo Kong:** Conceptualization, Methodology, Writing-review and editing, Project administration.

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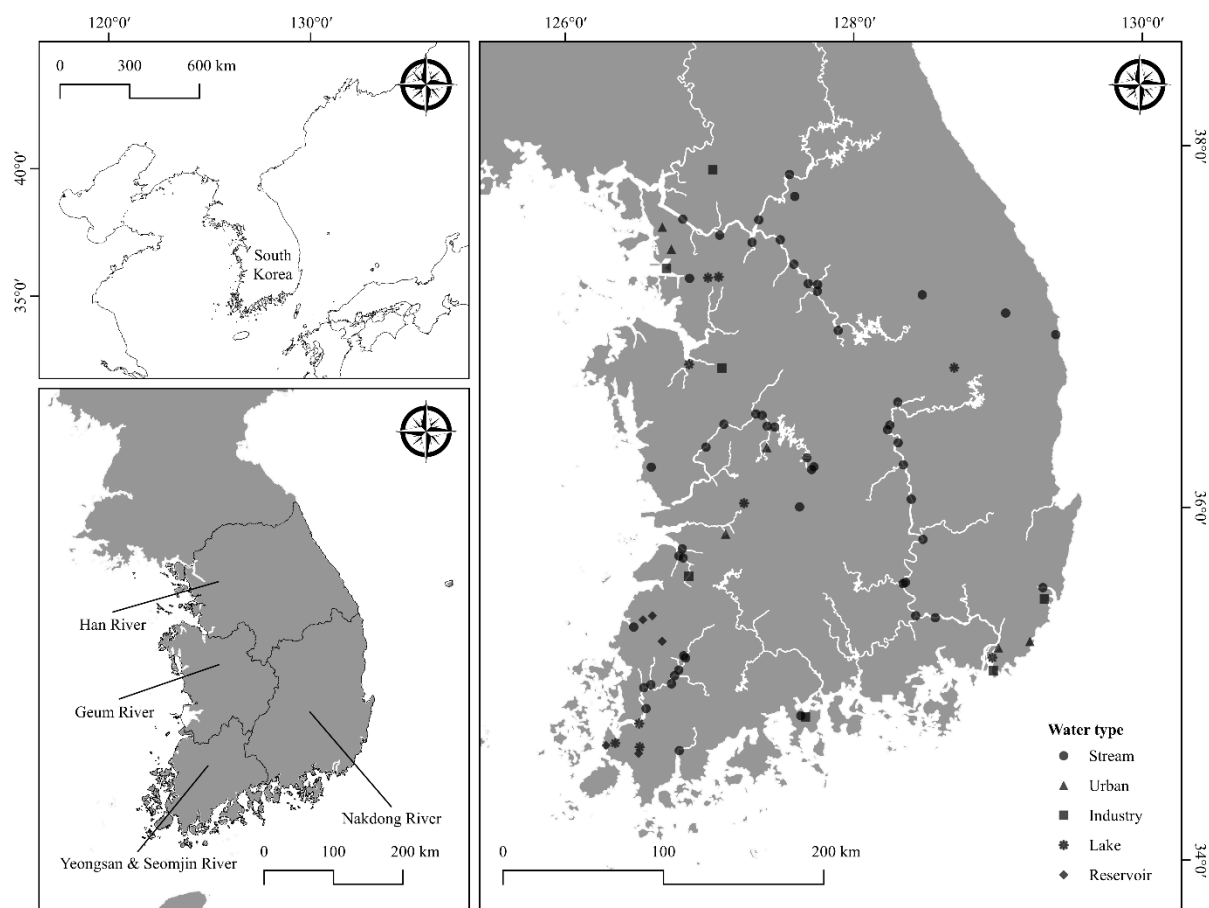


Figure 1. Distribution of the 80 freshwater sediment monitoring sites used for developing the Korean Benthic Macroinvertebrate Sediment Index (KBSI) across major river basins in Korea. Site categories indicate broad differences in waterbody type, surrounding land-use setting, and depositional sediment environment, rather than reference or impairment classes.

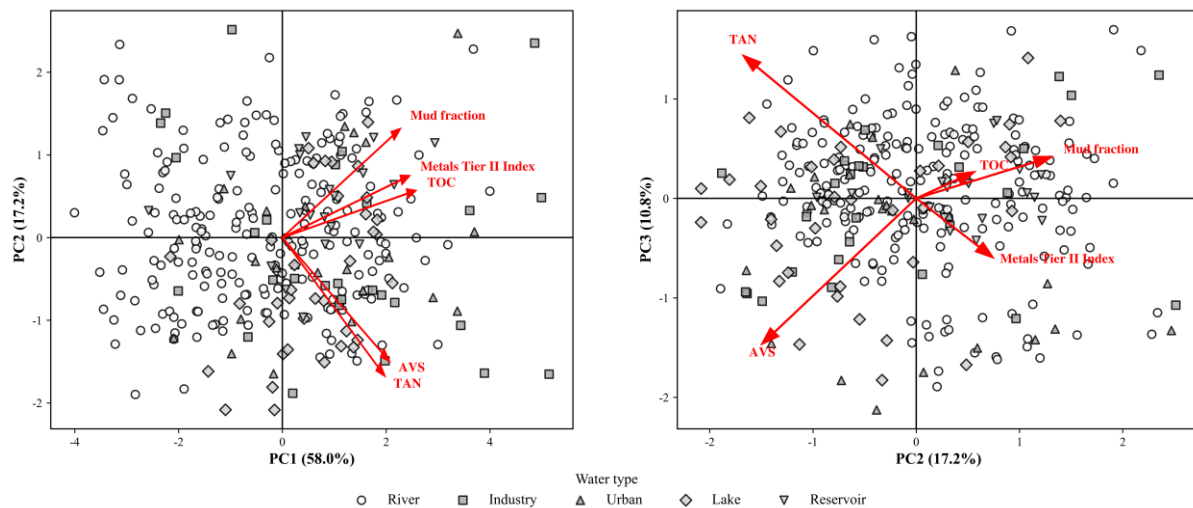


Figure 2. PCA biplots of sediment physicochemical variables. The left panel shows the PC1–PC2 ordination, and the right panel shows the PC2–PC3 ordination. Environmental vectors represent porewater total ammonia nitrogen (TAN), acid-volatile sulfides (AVS), total organic carbon (TOC), Metals Tier II Index, and mud fraction. PC1 explained 58.0% of the total variance and was used as the primary field-based sediment-condition gradient for deriving taxon-specific tolerance values and indicator weights. Site symbols indicate descriptive waterbody types and were not used as reference or impairment classes.

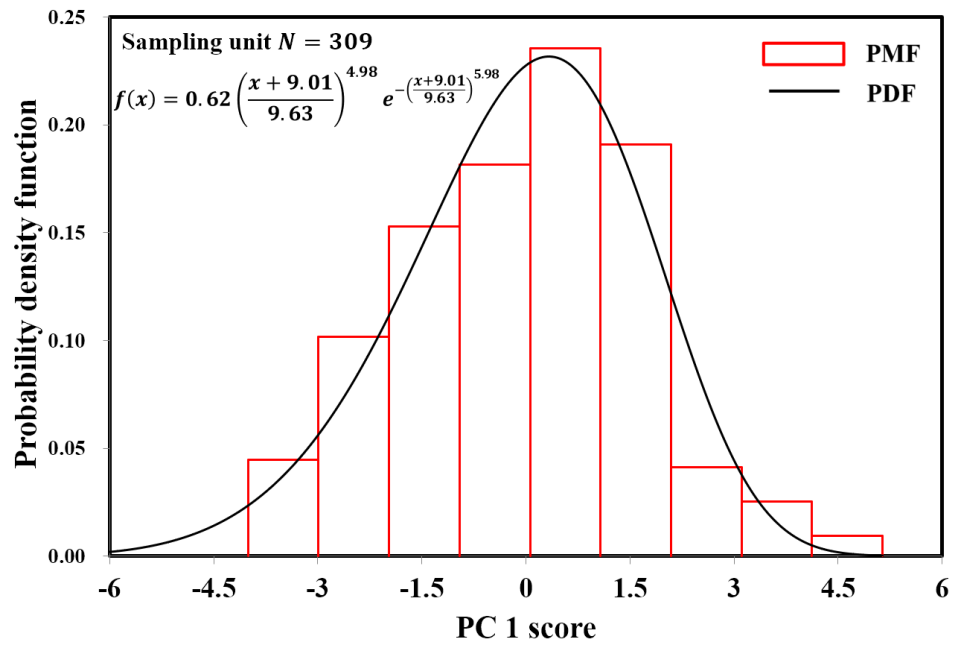
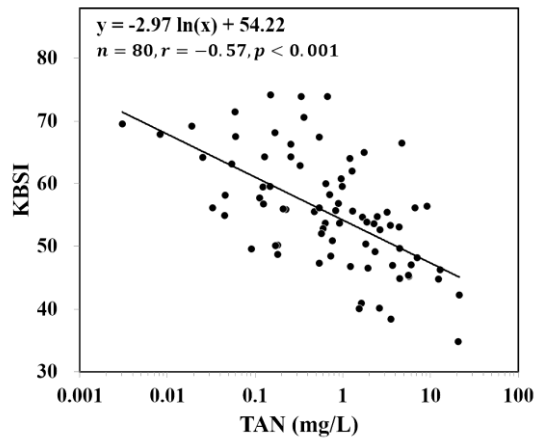
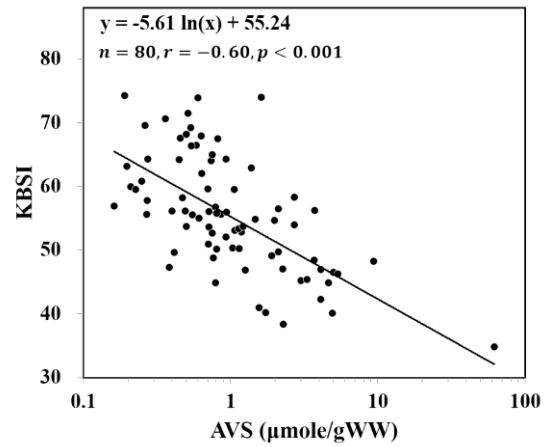


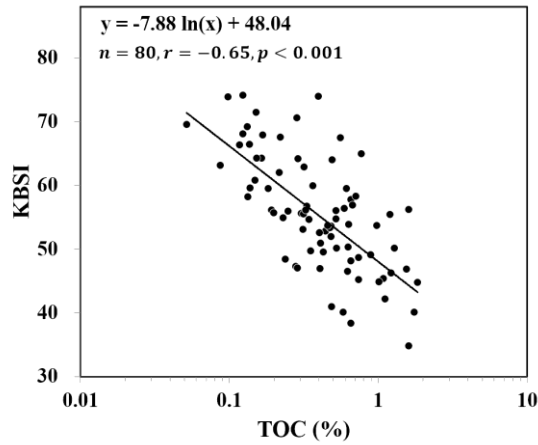
Figure 3. Probability distribution of PC1 scores for 309 sample units containing indicator taxa. The empirical distribution of PC1 scores was fitted using a Weibull cumulative distribution function. Vertical dashed lines indicate the nine class boundaries derived from equal cumulative-probability intervals. These PC1 classes were used as empirical intervals of the sediment-condition gradient for deriving taxon-specific tolerance values and indicator weights.



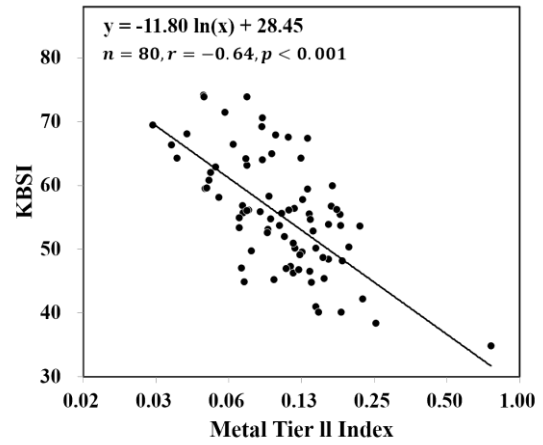
(a)



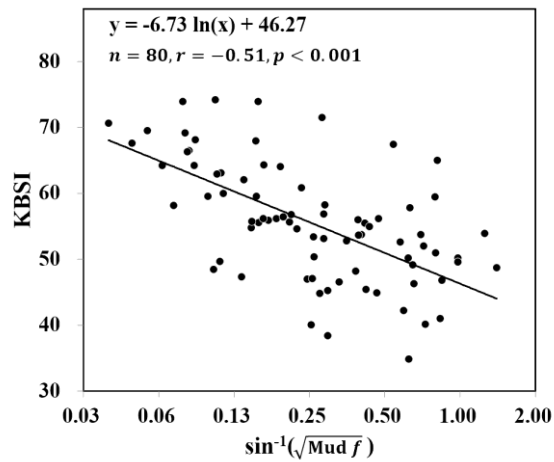
(b)



(c)



(d)



(e)

Figure 4. Relationships between KBSI and sediment environmental variables. Scatterplots show relationships between KBSI and (a) porewater total ammonia nitrogen (TAN), (b) acid-volatile sulfides (AVS), (c) total organic carbon (TOC), (d) Metals Tier II Index, and (e) mud fraction. Regression lines summarize field-level associations. KBSI showed significant negative relationships with all five sediment variables.

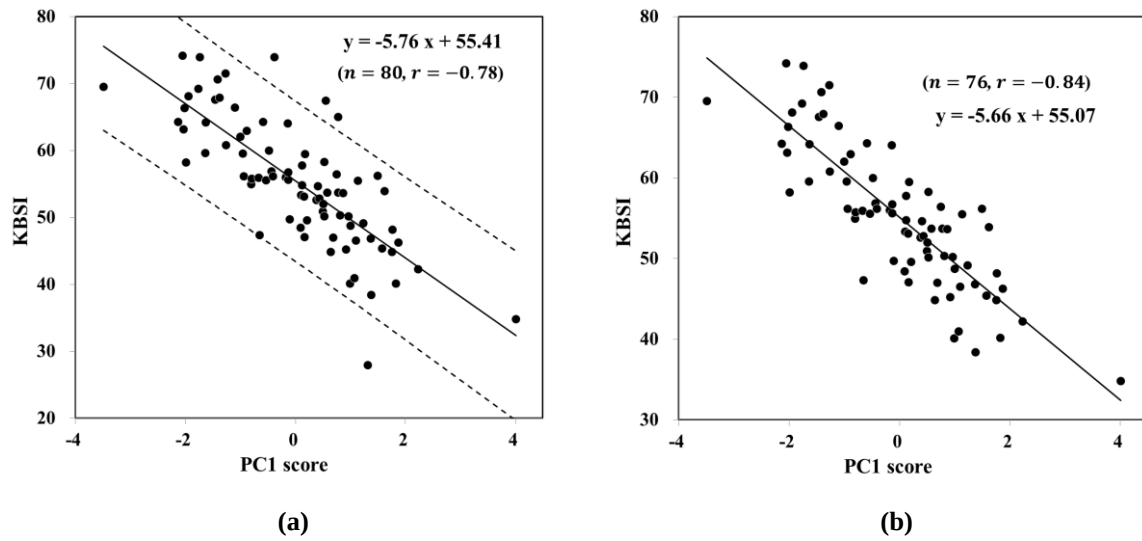


Figure 5. Relationship between PC1 score and KBSI. Scatterplots show the relationship between site-level annual mean PC1 scores and KBSI values before trimming (a) and after trimming four sites outside the 95% prediction interval (b). Higher PC1 scores indicate stronger sediment-associated environmental stress. KBSI showed a significant negative relationship with PC1 score, supporting the internal consistency of KBSI with the PCA-derived sediment-condition gradient.

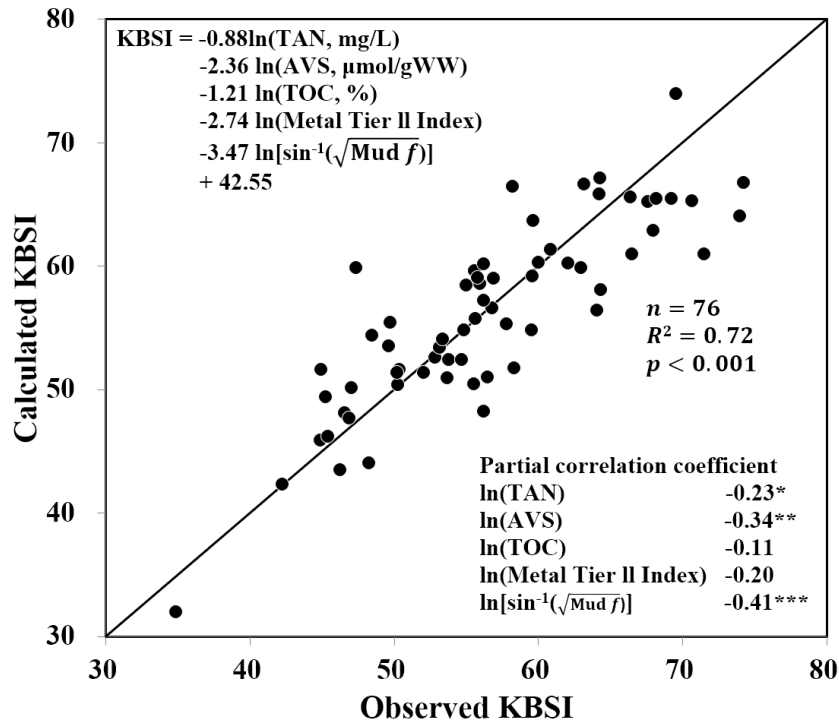


Figure 6. Observed and calculated KBSI values. The scatterplot shows the relationship between observed KBSI and calculated KBSI for 76 sites after excluding four sites outside the 95% prediction interval. The relationship indicates internal agreement within the index-development framework and should not be interpreted as external validation.

Table 1. PCA loadings and explained variance of sediment physicochemical variables.

Variable	PC1	PC2	PC3
TAN	0.66	0.03	0.00
AVS	0.69	−0.01	−0.06
TOC	0.86	0.08	−0.02
Metals Tier II Index	0.82	−0.01	−0.03
Mud fraction	0.76	−0.03	0.01
Variance explained (%)	58.0	17.2	10.8
Cumulative variance explained (%)	58.0	75.2	86.0
Eigenvalue	2.9	0.9	0.5

* TAN, total ammonia nitrogen; AVS, acid-volatile sulfides; TOC, total organic carbon. Mud fraction refers to the silt–clay fraction and was treated as a physical depositional variable rather than a chemical contaminant.

Table 2. PC1 class intervals and representative class values

Class No.	Cumulative probability	PC1 class interval	Representative class value
1	1/9	< -2.277	-3.142
2	2/9	-2.277 to < -1.367	-1.822
3	3/9	-1.367 to < -0.730	-1.049
4	4/9	-0.730 to < -0.199	-0.465
5	5/9	-0.199 to < 0.288	0.045
6	6/9	0.288 to < 0.773	0.531
7	7/9	0.773 to < 1.301	1.037
8	8/9	1.301 to < 1.976	1.639
9	1	≥ 1.976	3.559

*PC1 class intervals were derived from the Weibull cumulative distribution fitted to PC1 scores of the 309 sample units containing selected indicator taxa. Representative class values were used to derive taxon-specific tolerance values.

Table 3. Taxon-specific tolerance values and indicator weights.

No.	Taxon	Tolerance value (t)	Indicator weight (g)
1	<i>Anodonta woodiana</i>	3	4
2	<i>Corbicula fluminea</i>	3.1	2
3	<i>Musculium lacustre</i>	3.9	4
4	<i>Hediste</i> sp.	6.5	2
5	<i>Branchiura sowerbyi</i>	5.1	1
6	<i>Bothrioneurum vejdoskynum</i>	3.7	1
7	<i>Rhyacodrilus coccineus</i>	0	5
8	<i>Tubifex tubifex</i>	4.6	1
9	<i>Limnodrilus claparedeianus</i>	0.7	4
10	<i>Limnodrilus hoffmeisteri</i>	4.9	1
11	<i>Limnodrilus udekimineanus</i>	7.3	5
12	<i>Haplotaxis gordioides</i>	3.8	1
13	<i>Chaetogaster diaphanus</i>	1.9	4
14	<i>Chaetogaster limnaei</i>	0	5
15	<i>Branchiodrilus hortensis</i>	7.1	3
16	<i>Opidonais</i> sp.	0	4
17	<i>Amphichaeta asiatica</i>	3.8	2
18	<i>Haemonais waldvogeli</i>	4.8	2
19	<i>Nais variabilis</i>	5.6	3
20	<i>Nais communis</i>	7.3	5
21	<i>Slavina appendiculata</i>	7.3	5
22	<i>Stylaria fossularis</i>	3.1	3
23	<i>Pristina biserrata</i>	0	5
24	<i>Pristina longiseta</i>	1.8	3
25	<i>Henlea</i> sp.	5.7	3
26	<i>Mesenchytreanus</i> sp.	6.9	2
27	<i>Lumbriculus variegatus</i>	3.3	1
28	<i>Lamprotus orientalis</i>	0.9	3
29	<i>Eisenia koreana</i>	4.2	1
30	<i>Helobdella stagnalis</i>	5.6	1
31	<i>Erpobdella lineata</i>	4.3	1
32	<i>Erpobdella</i> sp.	7.9	2
33	<i>Cyprididae</i> sp.	7.3	5
34	<i>Gnorimosphaeroma</i> sp.	4.6	5
35	<i>Cyathura higoensis</i>	7.9	4
36	<i>Asellus</i> sp.	6.4	2
37	<i>Gammarus</i> sp.	4.9	2
38	<i>Jesogammarus</i> sp.	10	5
39	<i>Monocorophium</i> sp.	9	5
40	<i>Collembola</i> sp.	4.2	2

41	<i>Baetis fuscatus</i>	3.3	4
42	<i>Cloeon dipterum</i>	5.1	2
43	<i>Labiobaetis atrebatinus</i>	4.6	2
44	<i>Procloeon maritimum</i>	0	5
45	<i>Procloeon pennulatum</i>	3.3	2
46	<i>Choroterpes altioculus</i>	0.8	3
47	<i>Potamanthus formosus</i>	0	4
48	<i>Rhoenanthus coreanus</i>	0.2	3
49	<i>Ephoron shigae</i>	0	5
50	<i>Ephemera orientalis</i>	2.7	2
51	<i>Serratella setigera</i>	5.6	2
52	<i>Caenis</i> sp.	2.6	2
53	<i>Psychoda</i> sp.	4	4
54	<i>Dixidae</i> sp.	3.3	5
55	<i>Ceratopogonidae</i> sp.	4.3	2
56	<i>Benthalia</i> sp.	4.6	5
57	<i>Chironomus circumdatus</i>	0	5
58	<i>Chironomus flaviplumus</i>	5	1
59	<i>Chironomus kiiensis</i>	8.5	3
60	<i>Chironomus nipponensis</i>	4.5	2
61	<i>Chironomus edwardsi</i>	6.9	3
62	<i>Chironomus</i> sp.	6	2
63	<i>Cladotanytarsus vanderwulpi</i>	3.3	3
64	<i>Cryptochironomus</i> sp.	1.6	2
65	<i>Demicryptochironomus</i> sp.	1.7	3
66	<i>Dicrotendipes nervosus</i>	3.8	3
67	<i>Dicrotendipes pelochloris</i>	9	5
68	<i>Dicrotendipes septemmaculatus</i>	6.9	3
69	<i>Dicrotendipes</i> sp.	6.6	2
70	<i>Glyptotendipes tokunagai</i>	5.2	2
71	<i>Harnischia japonica</i>	4.9	3
72	<i>Harnischia gumsanea</i>	7.3	5
73	<i>Harnischia</i> sp.	6.2	2
74	<i>Lipiniella moderata</i>	4.5	2
75	<i>Microchironomus tener</i>	3.1	3
76	<i>Microchironomus</i> sp.	7.3	5
77	<i>Microtendipes</i> sp.	3.8	3
78	<i>Nilothauma</i> sp.	3.9	4
79	<i>Parachironomus gracilior</i>	1.7	5
80	<i>Parachironomus arcuatus</i>	7.4	3
81	<i>Paratanytarsus inopertus</i>	7.3	5
82	<i>Paratanytarsus grimmii</i>	7.9	4
83	<i>Paratanytarsus</i> sp.	4.6	5

84	<i>Paratendipes albimanus</i>	2.2	2
85	<i>Paratendipes</i> sp.	0	4
86	<i>Polypedilum asakawaense</i>	6.6	2
87	<i>Polypedilum pedestre</i>	0	5
88	<i>Polypedilum nubifer</i>	1.3	3
89	<i>Polypedilum decematoguttatum</i>	4.6	5
90	<i>Polypedilum masudai</i>	4	2
91	<i>Polypedilum cultellatum</i>	3.8	2
92	<i>Polypedilum nubeculosum</i>	5.5	2
93	<i>Polypedilum</i> sp.	4.7	1
94	<i>Stictochironomus sinsauensis</i>	3	1
95	<i>Stictochironomus</i> sp.	4.8	3
96	<i>Tanytarsus kiseogi</i>	0.8	4
97	<i>Tanytarsus takahashii</i>	1.7	5
98	<i>Tanytarsus tamagotoi</i>	3.3	5
99	<i>Tanytarsus tamakutibasi</i>	2.3	4
100	<i>Tanytarsus formosanus</i>	10	5
101	<i>Tanytarsus</i> sp.	4.3	1
102	<i>Pothastia</i> sp.	4.6	5
103	<i>Thienemanniella</i> sp.	1.2	4
104	<i>Brillia japonica</i>	4.6	5
105	<i>Corynoneura</i> sp.	1.7	3
106	<i>Cricotopus bicinctus</i>	2.7	3
107	<i>Cricotopus bimaculatus</i>	0	4
108	<i>Cricotopus sylvestris</i>	5.8	1
109	<i>Cricotopus triannulatus</i>	2.3	4
110	<i>Cricotopus</i> sp.	4.3	1
111	<i>Eukiefferiella</i> sp.	0	4
112	<i>Hydrobaenus kondoi</i>	3.9	1
113	<i>Hydrobaenus</i> sp.	3.3	5
114	<i>Limnophyes tamakitanaides</i>	6	5
115	<i>Orthocladius glabripennis</i>	5.6	4
116	<i>Orthocladius</i> sp.	1.8	2
117	<i>Paratrichocladius rufiventris</i>	6.1	2
118	<i>Prosilocerus akamusi</i>	2.3	4
119	<i>Rheocricotopus</i> sp.	0	5
120	<i>Ablabesmyia longistyla</i>	8.3	3
121	<i>Ablabesmyia monilis</i>	3.7	2
122	<i>Ablabesmyia prorasha</i>	0.8	3
123	<i>Ablabesmyia</i> sp.	4.5	1
124	<i>Conchapelopia</i> sp.	3.4	1
125	<i>Hayesomyia tripunctata</i>	9	5
126	<i>Procladius choreus</i>	2.9	2

127	<i>Procladius nigriventris</i>	10	5
128	<i>Procladius</i> sp.	5.1	1
129	<i>Rheopelopia</i> sp.	3.3	5
130	<i>Tanypus punctipennis</i>	9.3	3
131	<i>Saetheria tylus</i>	0	4
132	<i>Cladopelma edwardsi</i>	7	2
133	<i>Cladopelma virescens</i>	3.3	4
134	<i>Chironomidae</i> sp.	1.8	2
135	<i>Syrphidae</i> sp.	7.3	5
136	<i>Ephydridae</i> sp.	10	4
137	<i>Ecnomus</i> sp.	7.6	2

*Tolerance values represent the weighted average position of each taxon along the nine PC1 class intervals. Indicator weights represent the concentration of each taxon's occurrence distribution along the PC1 gradient and should not be interpreted as sensitivity values.

Table 4. KBSI classification criteria for sediment-associated ecological condition

Class	Ecological condition	KBSI range
A	High	70–100
B	Good	60–<70
C	Moderate	50–<60
D	Poor	40–<50
E	Bad	0–<40

*KBSI classes were defined as empirical biological assessment categories for sediment-associated ecological condition. The classes should not be interpreted as independent chemical sediment-quality standards or toxicological thresholds.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: